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ADM NO: BSCCS/2024/52266

COURSE: BACHELOR OF SCIENCE IN COMPUTER SCIENCE

PROJECT TITLE: WAN SHOES INVENTORY MANAGEMENT SHOES

DEPARTMENT: COMPUTING AND INFORMATICS

LECTURER : MICHAEL NYORO

GITHUB LINK: https://github.com/w4nlouis/wan-shoes-repository

Contents

WEEK 1 CHECKLIST

Fig 1.1

XAMPP Control Panel showing Apache and MySQL services running with green indicators.

Figure 1XAMPP Control Panel

Fig 1.2:

Browser accessing http://localhost/phpmyadmin displaying phpMyAdmin dashboard. This confirms MySQL server is running and accessible via web interface.

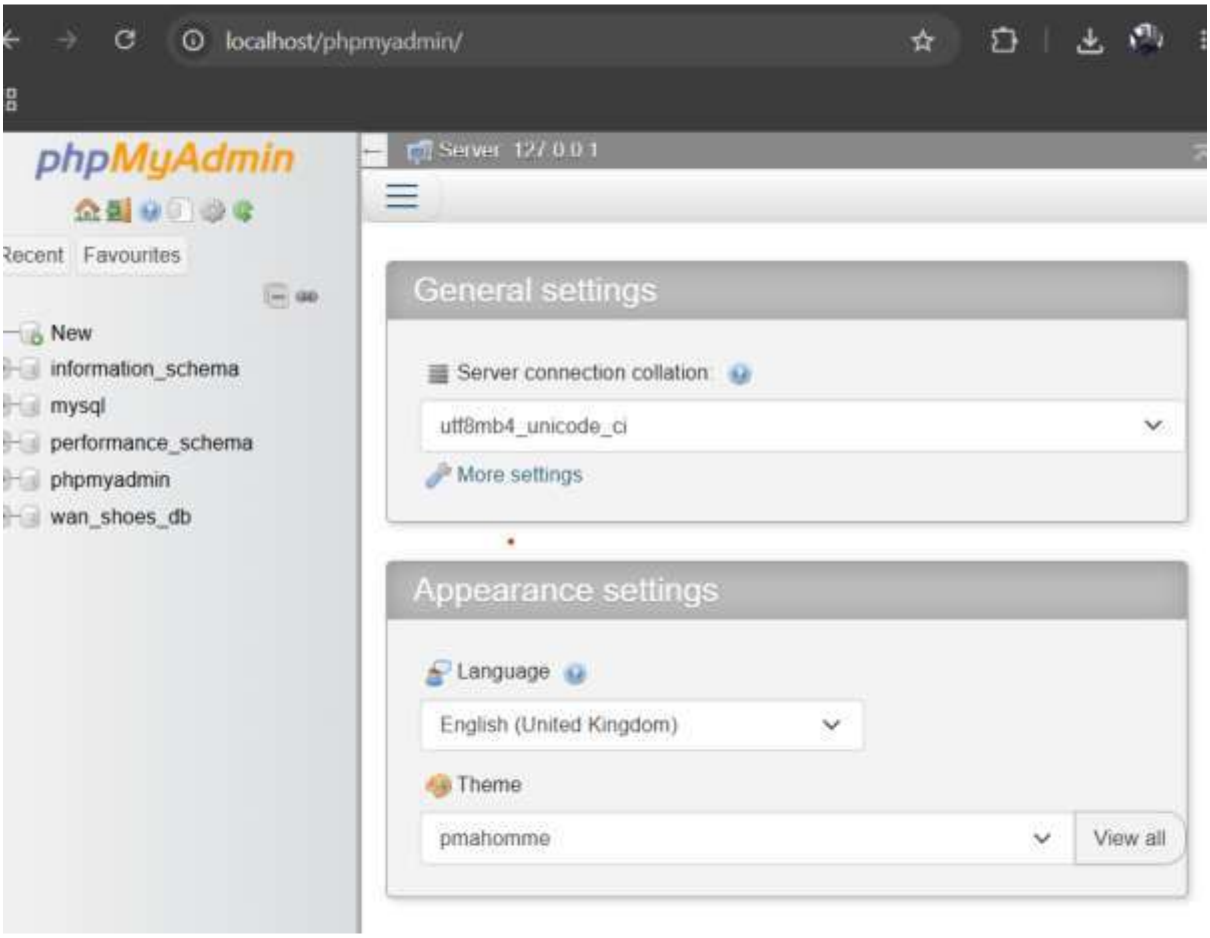


Figure 2phpmyadmin interface

Fig 1.3

Test page http://localhost/wan_shoes/index.php outputting "Hello World" from PHP. The <?php echo "Hello World from PHP"; ?> script executed successfully, confirming PHP processing is working.

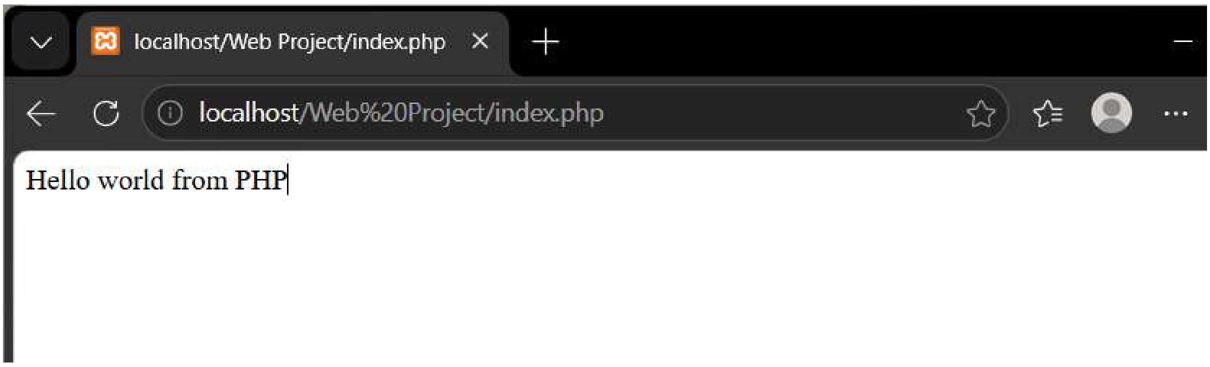


Figure 3Proof of PHP working

Fig 1.4

PhpMyAdmin interface showing wan_shoes_db database created with utf8_general_ci collation. The database contains five tables: admins, employees, products, sales, and stock_records

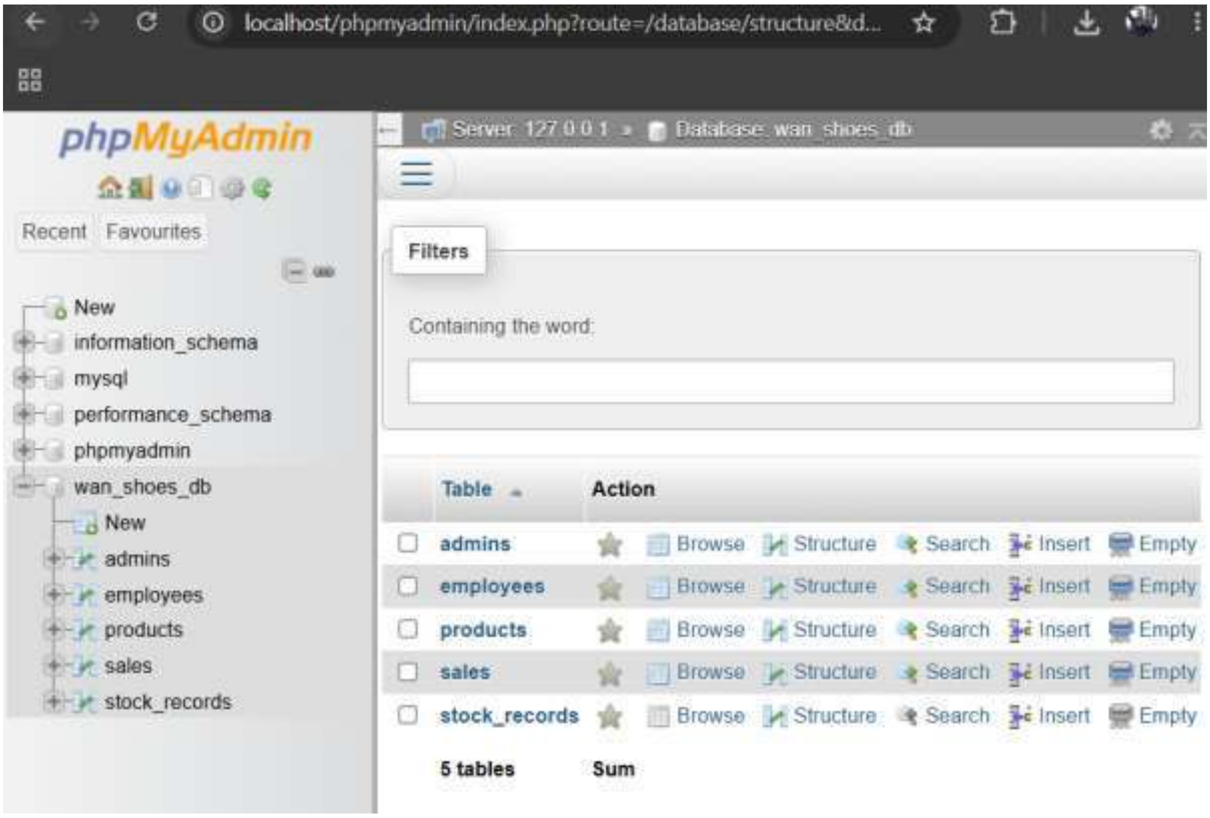


Figure 4 wan_shoes_d

Fig 1.5

S Code folder structure showing wan_shoes project root with subdirectories: admin/, css/, images/, js/, products/, sales/, stock/, and includes/. This follows professional web development organization standards.

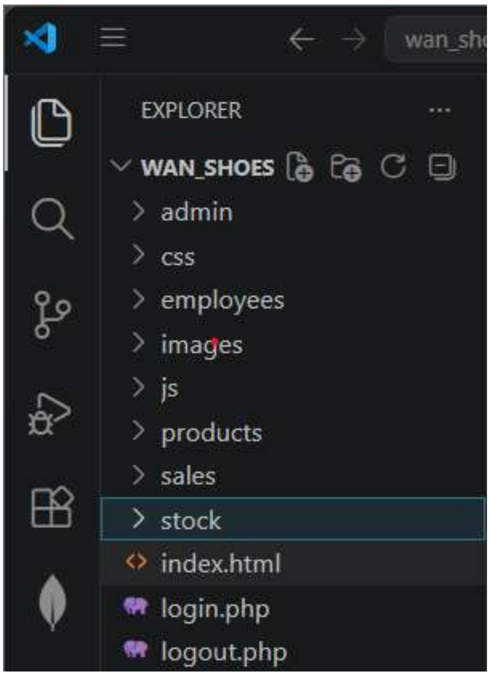
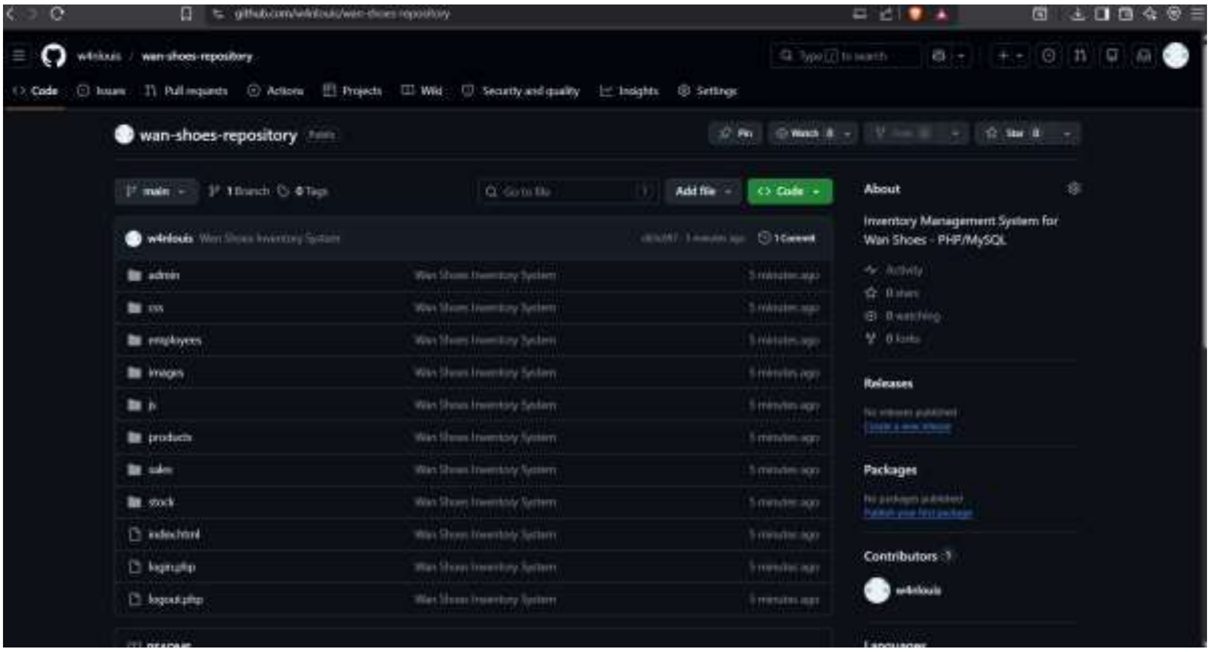


Figure 5 Folder Organization

Fig 1.6

GitHub repository at https://github.com/w4nlouis/wan-shoes-repository showing all project files uploaded including PHP scripts, CSS styling, JavaScript validation, and database export file.



Week 1 Student Reflection

I installed XAMPP on Windows 11 to create a local web development environment for the Wan Shoes inventory system. Fig 1.1 shows Apache and MySQL services running without port conflicts, which is essential for PHP processing and database connectivity. Fig 1.3 demonstrates successful PHP execution where a simple echo statement rendered "Hello World" in the browser, confirming the server properly parses PHP code. Fig 1.4 displays wan_shoes_db created in phpMyAdmin with five relational tables: admins, employees, products, sales, and stock_records. The folder structure in Fig 1.5 follows MVC-inspired organization with separate directories for CSS, JavaScript, and feature modules. Finally, Fig 1.6 shows the GitHub repository containing all project code, fulfilling the version control requirement. XAMPP provides a stable foundation for developing the complete inventory management system.

WEEK 2

Fig 2.1

Digital wireframe of the login page showing email field, password field, and sign-in button centered on the page. The design uses a clean card layout with brand title "WAN SHOES" at the top.

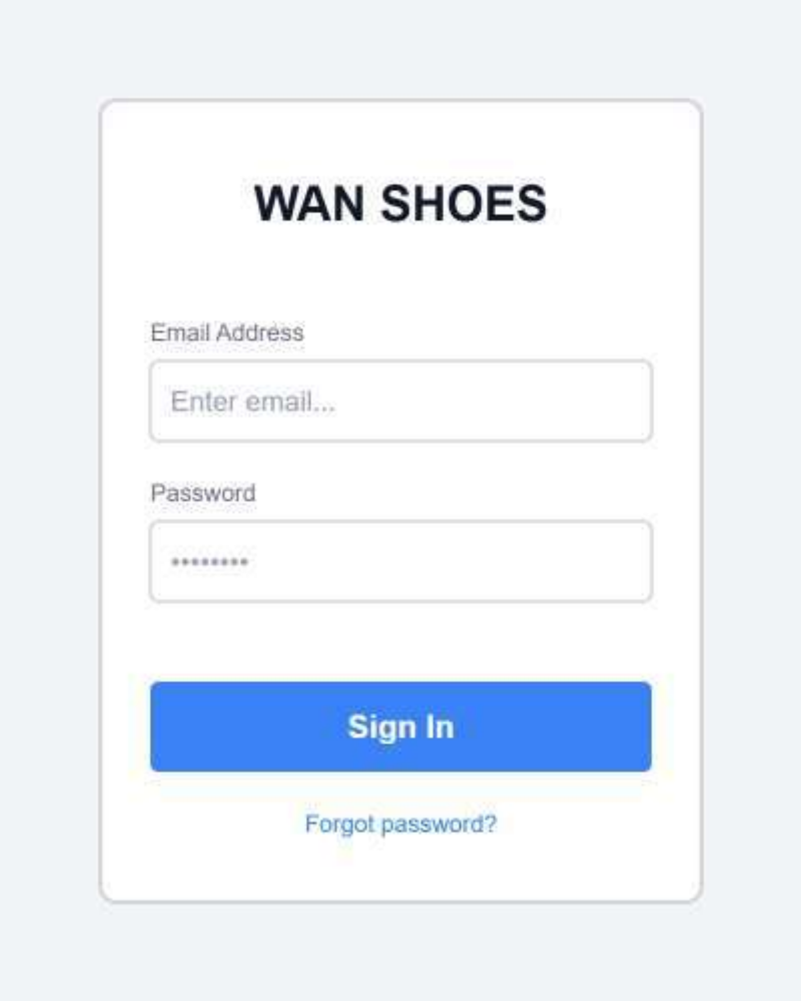


Figure 6Login wirefreame

Fig 2.2

Wireframe of the admin dashboard displaying three statistics cards (Products Count, Inventory Value, Staff Count) and three action buttons (Manage Products, Manage Employees, View Sales). Navigation follows a vertical card layout.

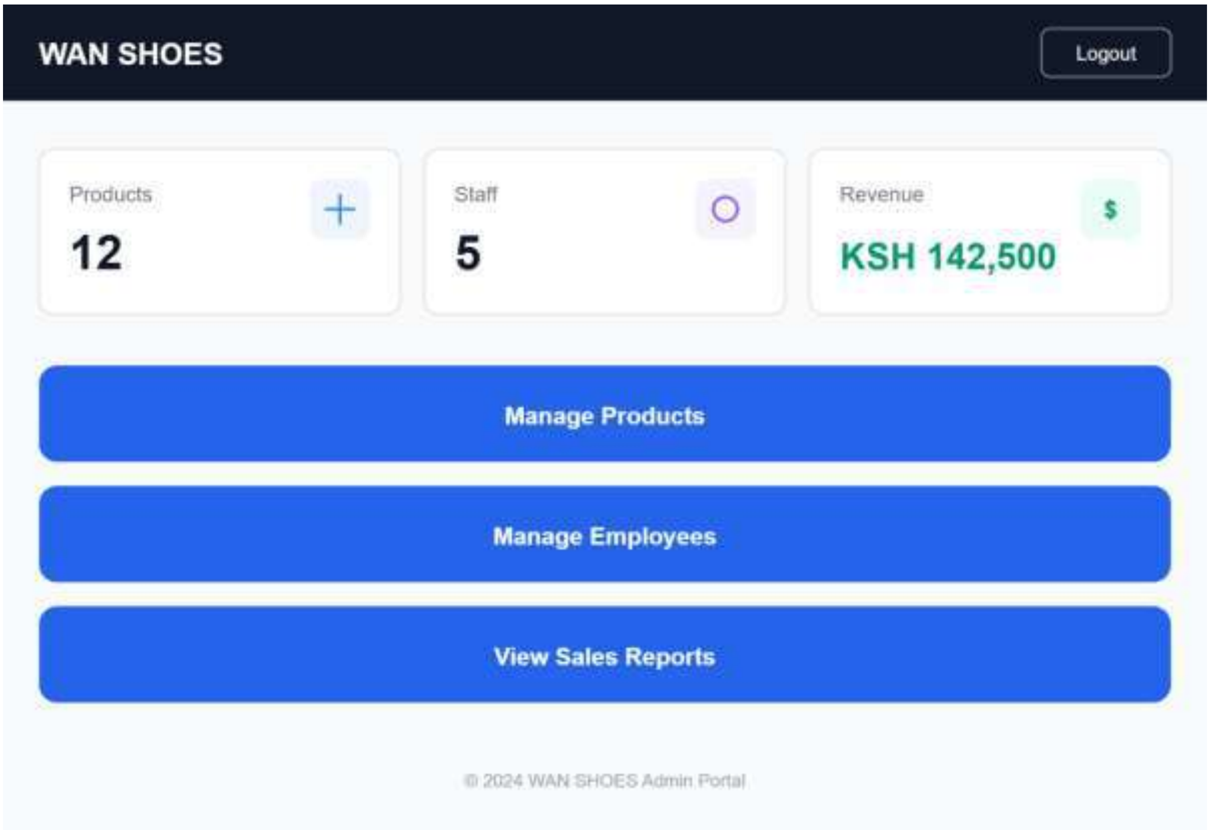


Figure 7 Dashboaard

Fig 2.3

Wireframe of the products management page for admin role showing an "Add New Product" form with fields for name, brand, size, price, and stock quantity, followed by a data table displaying existing products with edit and delete actions.

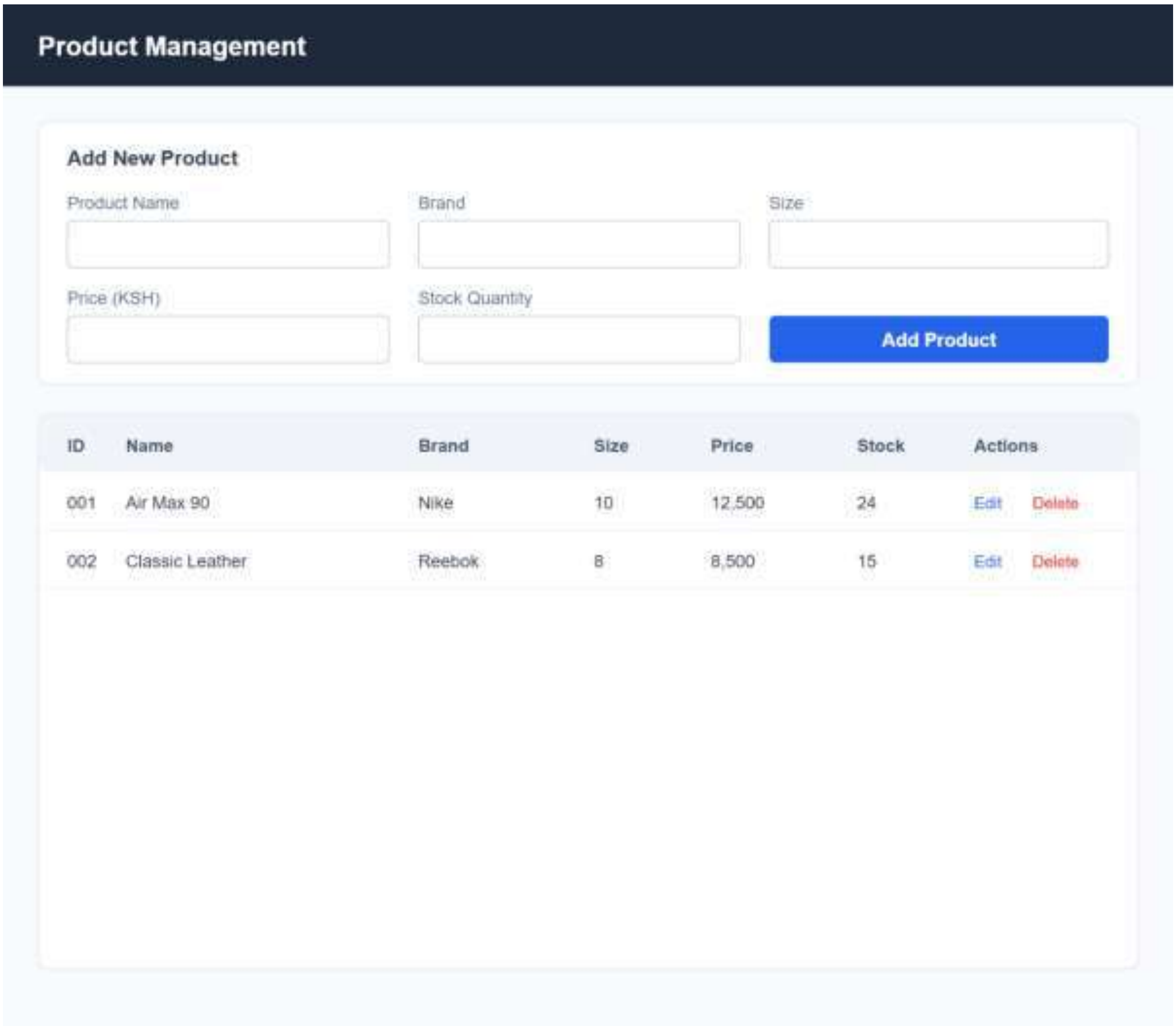


Figure 8 Products wireframe

FIG 2.4

GUI COLOR SCHEME

Element	**Color**	**Usage**
Primary Background	Dark Black	Body background
Card Background	White	Forms, containers, dashboards
Accent Color	Gold	Borders, hover effects, highlights
Button Primary	Black	Normal buttons
Button Danger	Red	Logout, delete actions
Text Primary	Black	Headings, labels
Text Secondary	Gray	Subtitles, helper text

WEEK 2 REFLECTION

Week 2 focused on UI/UX planning and system design for Wan Shoes inventory system. I created five digital wireframes covering login, admin dashboard, products page, sales recording, and sales reports. These wireframes establish the user flow for three roles: admin manages products and employees, clerks record sales, and stock managers add inventory. Fig 2.4 documents the GUI color scheme using black, white, and gold for a professional premium aesthetic. The project proposal summary defines technical specifications including PHP/MySQL stack and five database tables

Project Proposal Summary

Project Title: Wan Shoes Inventory Management System

Target Users: Internal staff only — Administrators, Clerks, and Stock Managers. No customer-facing module.

Purpose: Track shoe products, manage inventory levels, record sales transactions, monitor stock additions, and generate sales reports with revenue totals.

Tech Stack: PHP, MySQL, HTML5, CSS3, JavaScript NE, XAMPP local server, GitHub for version control.

Key Features:

- Role-based authentication (Admin, Clerk, Stock Manager)
- Product CRUD operations (Create, Read, Update, Delete)
- Employee account management (Admin only)
- Sales recording with automatic stock deduction (Clerk only)
- Stock addition with audit trail (Stock Manager only)
- Sales reports with revenue calculation (Admin only)
- Password strength validation
- Responsive design supporting mobile and desktop

Database Tables: admins, employees, products, sales, stock_records

Expected Outcomes: A fully functional internal inventory system that allows staff to manage shoes, process sales, track stock, and view sales analytics in real-time.

WEEK 3

Javascript and PHP Basics

<u>Javascript form validation</u>

Under the login page empty email or password is prevented

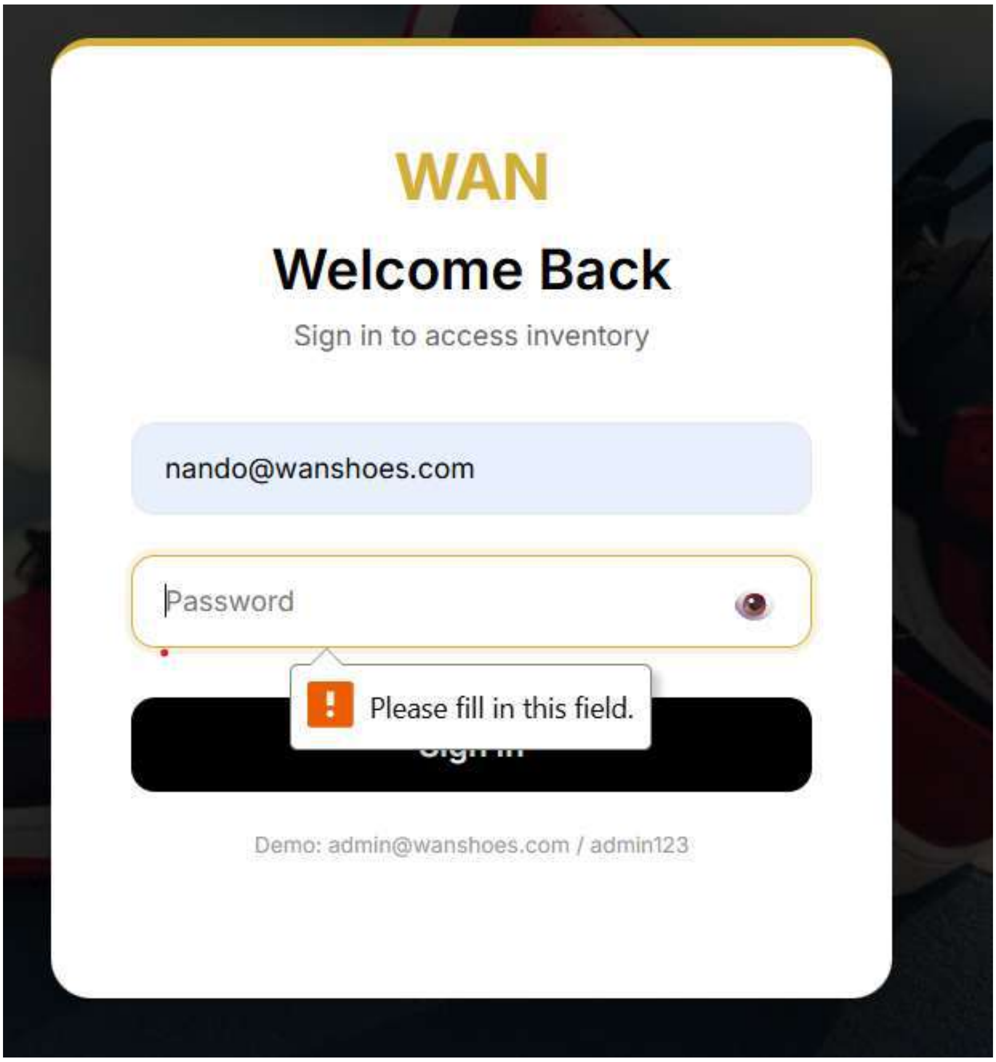


Figure 9 Form validation

<u>Password strength checker</u>

While hiring new employees the employee creation form gives feedback in real time concerning password strength

Figure 10 Strong password

Figure 11 Weak Password

<u>DOM Manipulation</u>

While using javascript I enabled one to be able to see their password while logging in

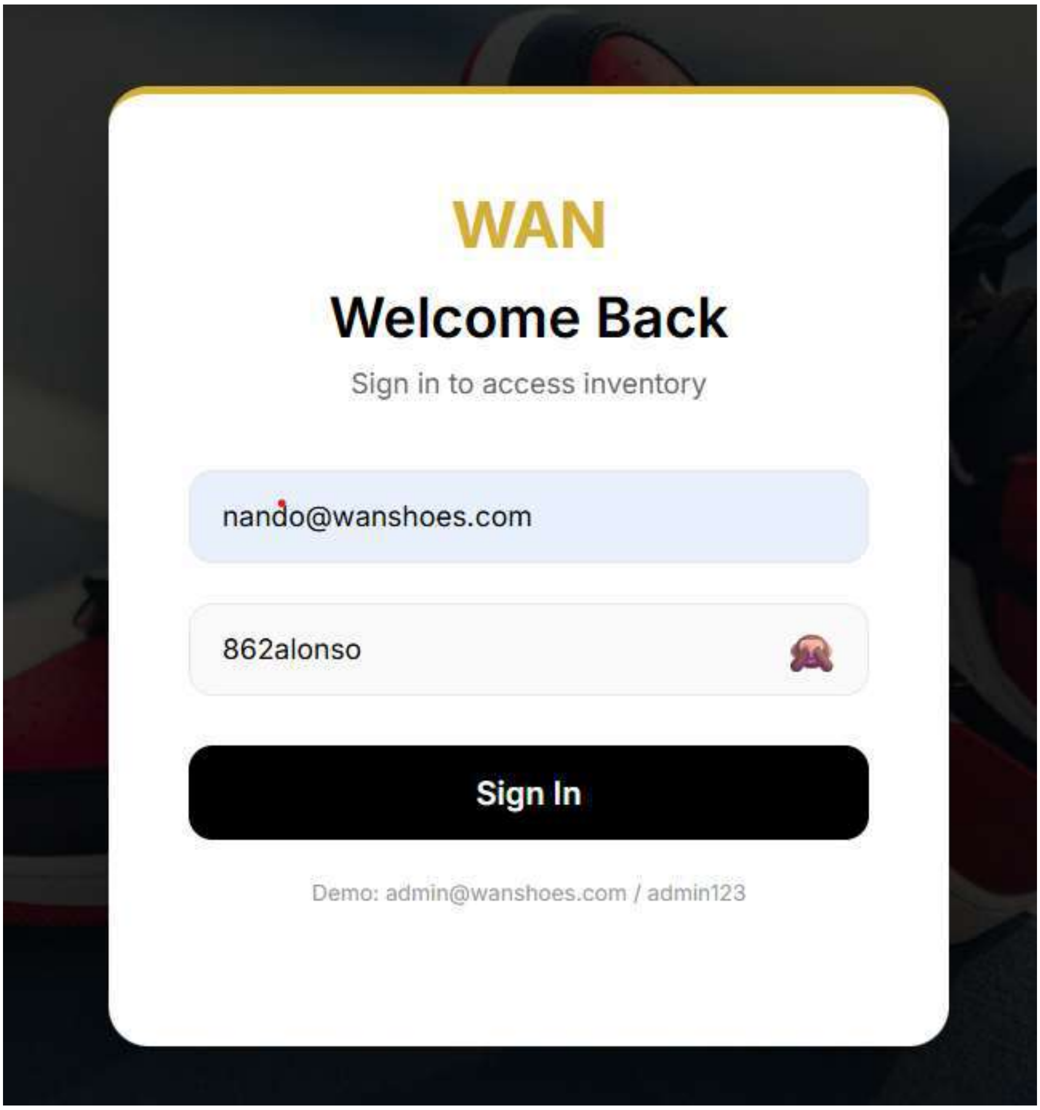


Figure 12 Password visible

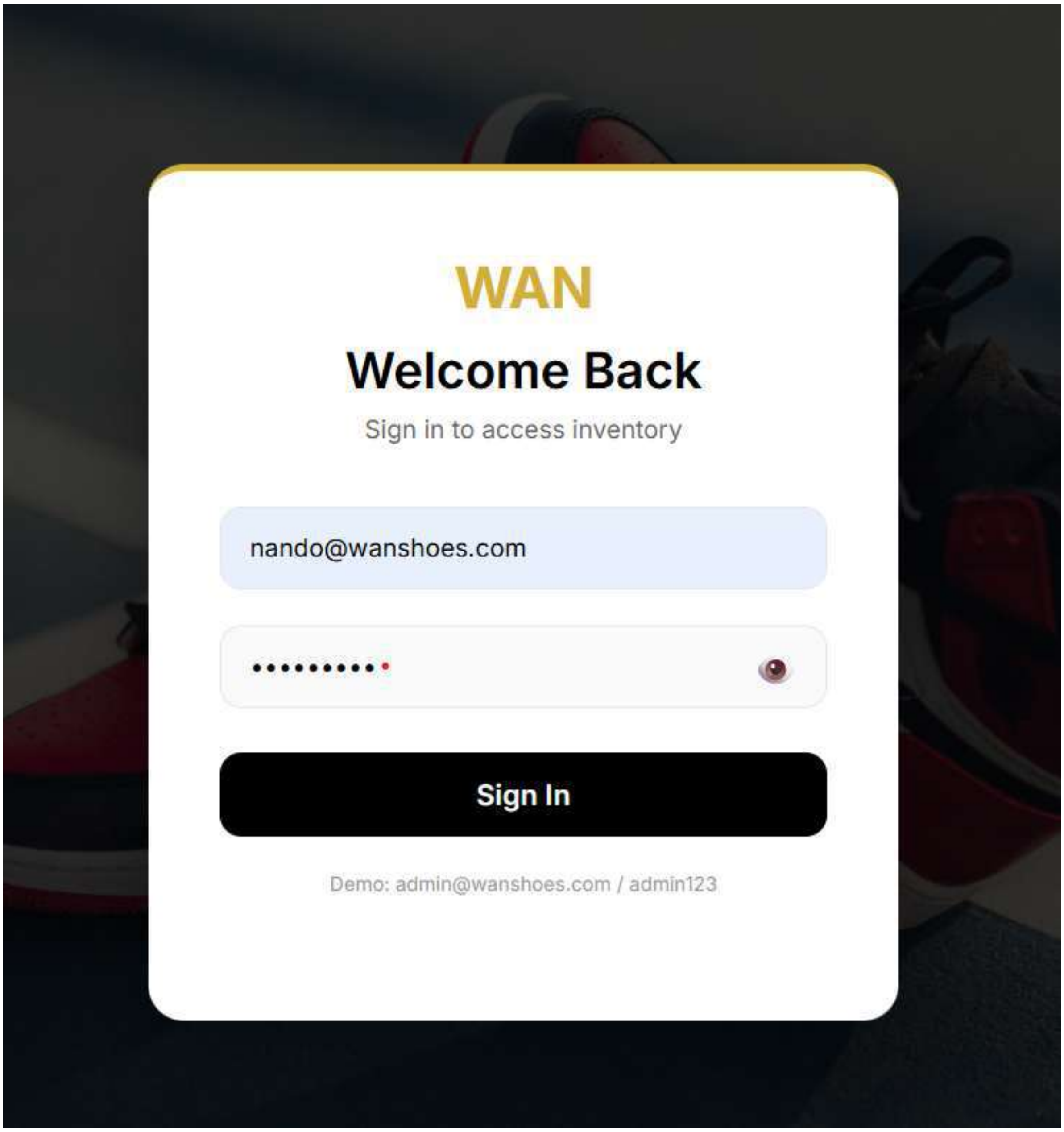


Figure 13Password hidden

Student Reflection

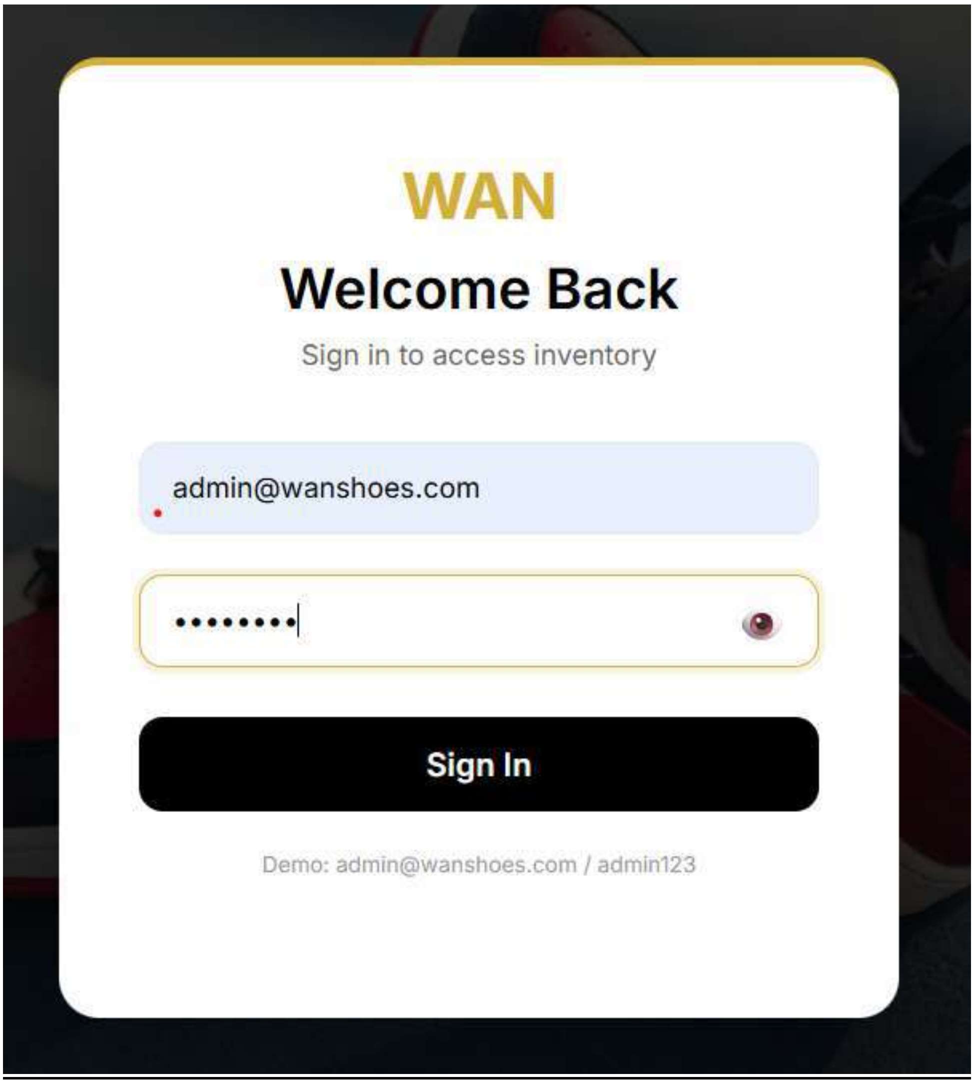
Week 3 introduced frontend validation using JavaScript and backend foundations with PHP for the Wan Shoes inventory system. I implemented client-side form validation on the login page using validateLogin() function, preventing empty email or password submissions before data reaches the server — improving user experience and reducing invalid requests. A real-time password strength checker was added to the employee creation form, evaluating passwords based on character length (4-7 chars = Medium, 8+ = Strong) and displaying color-coded feedback (red, orange, green). PHP syntax was verified using echo statements and variable manipulation in test files. Database connectivity was confirmed using mysqli_connect("localhost", "root", "", "wan_shoes_db"), successfully connecting to the MySQL database. A password visibility toggle using DOM manipulation was also implemented, allowing users to show/hide password text by clicking an eye icon. These foundational skills enable form processing, user authentication, and database integration for Weeks 4-5.

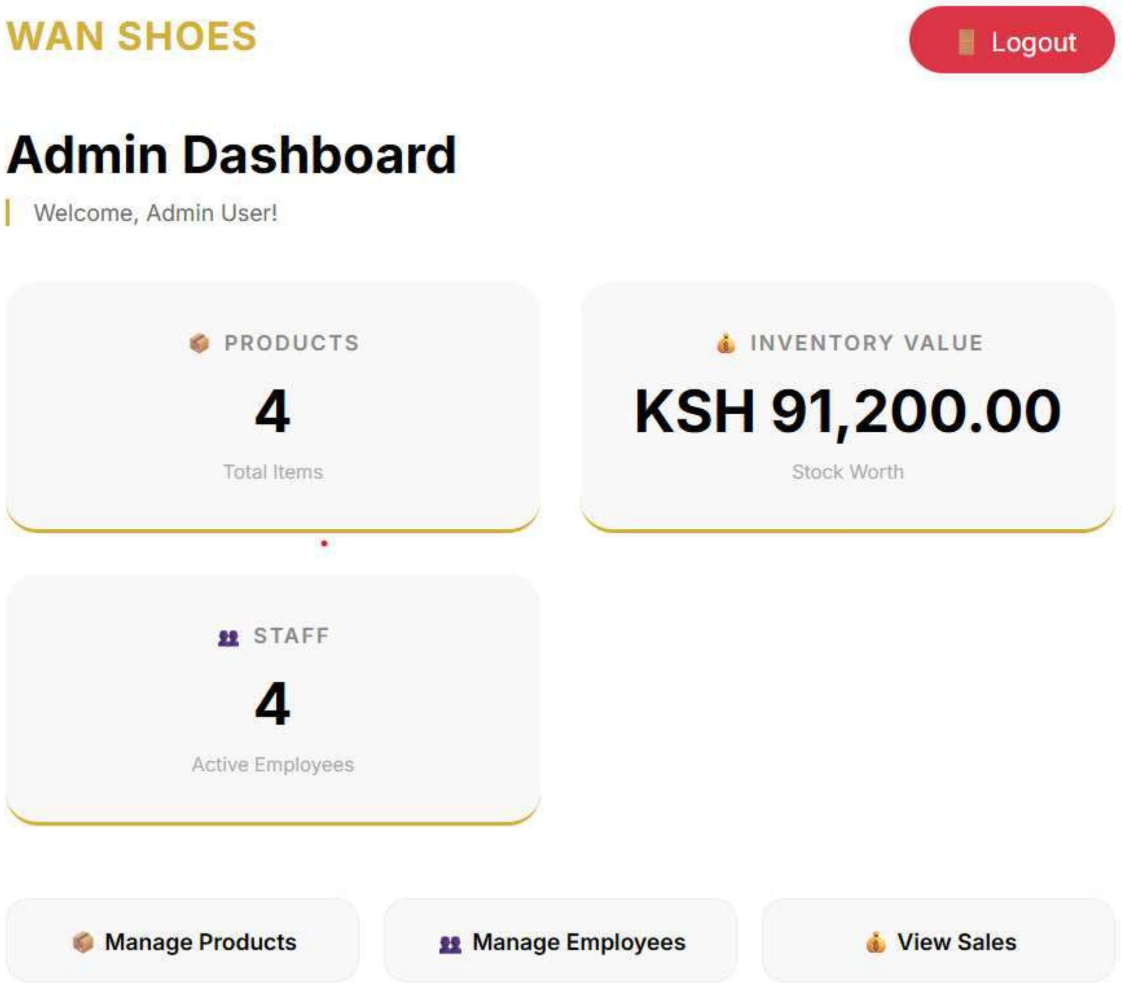
WEEK 4 Server Side Components and Backend Development

Role Based Login system

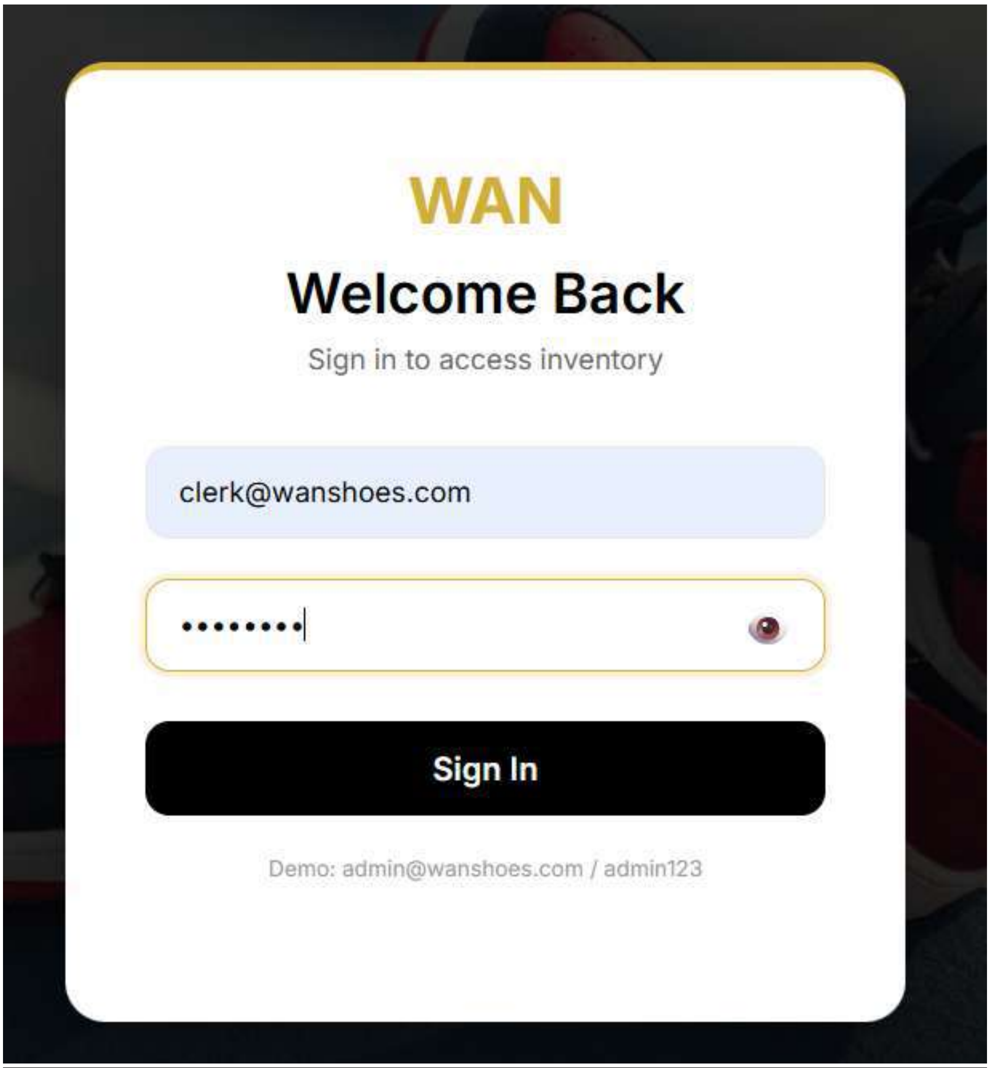
The system serves three roles of users under it ; admins, clerks and stock managers and their respective area of work is distinguished based on their login info

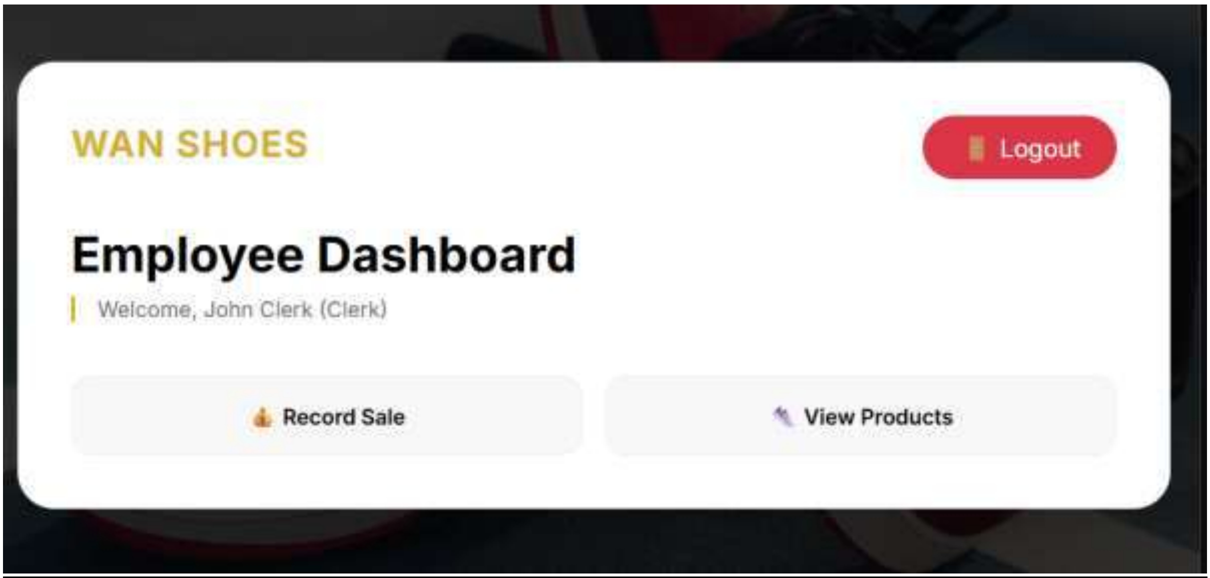
Admin login



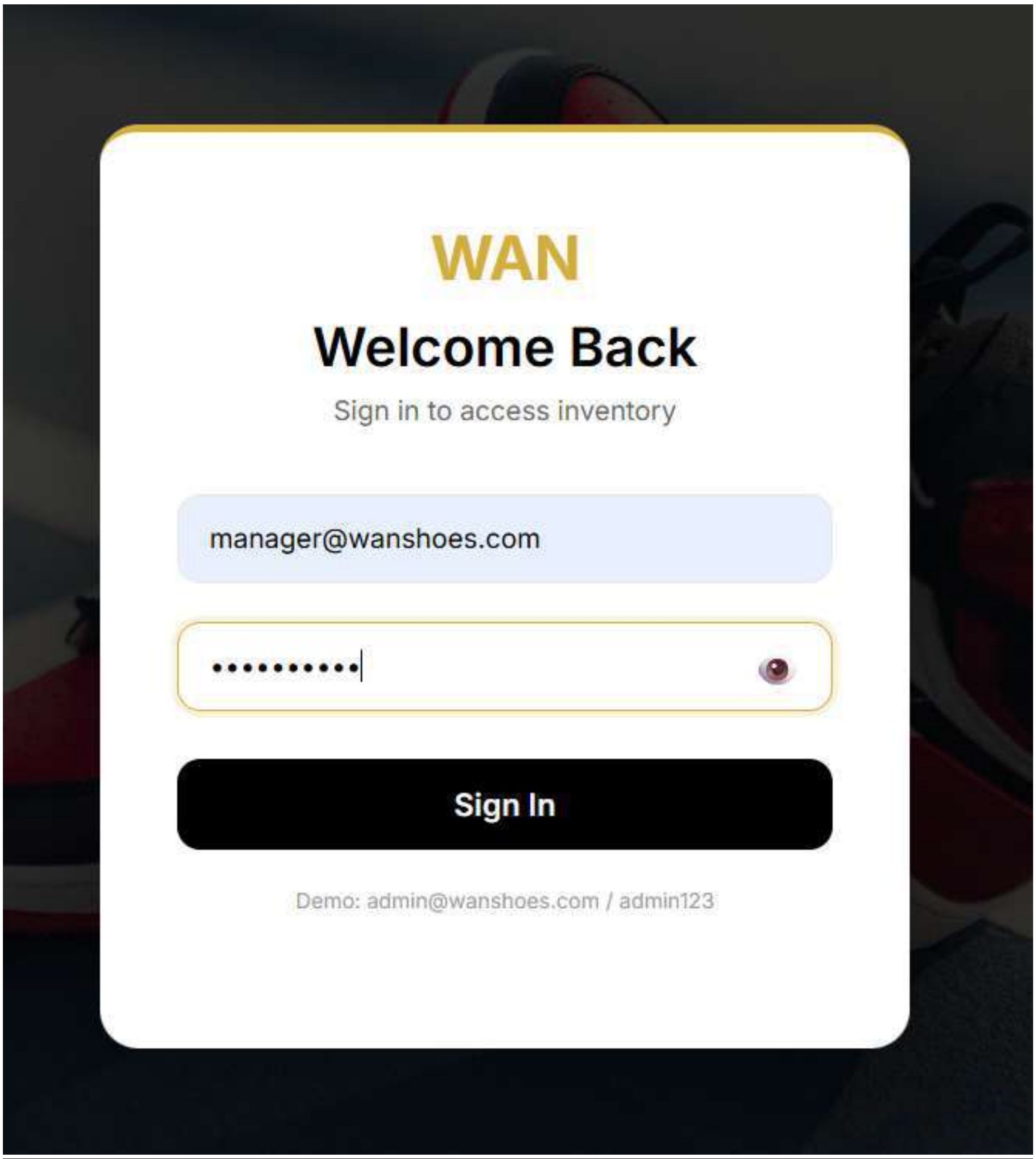

WAN SHOES
Logout
Admin Dashboard
Welcome, Admin User!
PRODUCTS
4
Total Items
INVENTORY VALUE
KSH 91,200.00
Stock Worth
STAFF
4
Active Employees
Manage Products
Manage Employees
View Sales

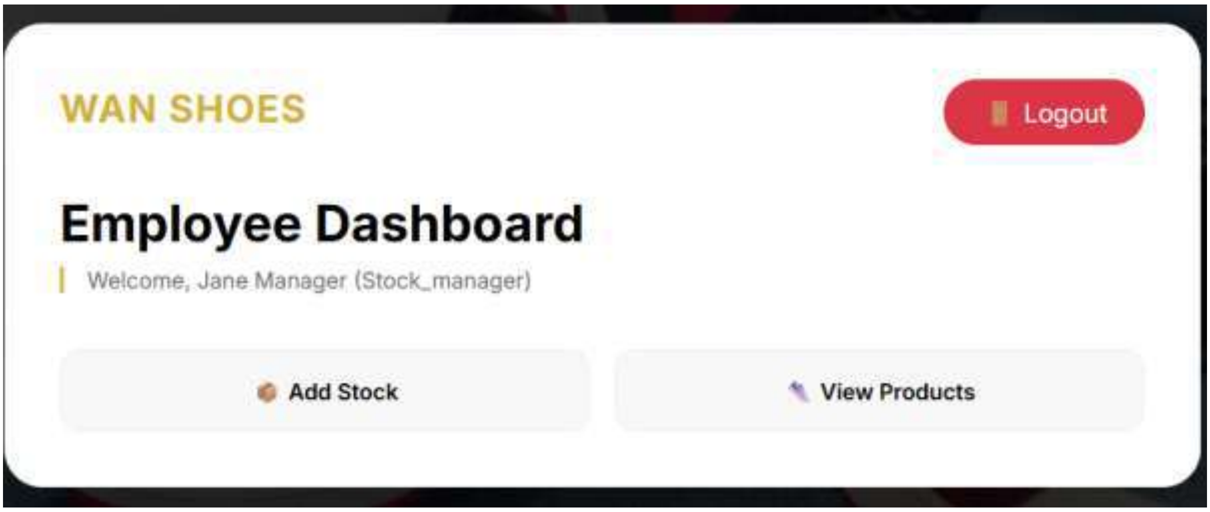
Clerk Login



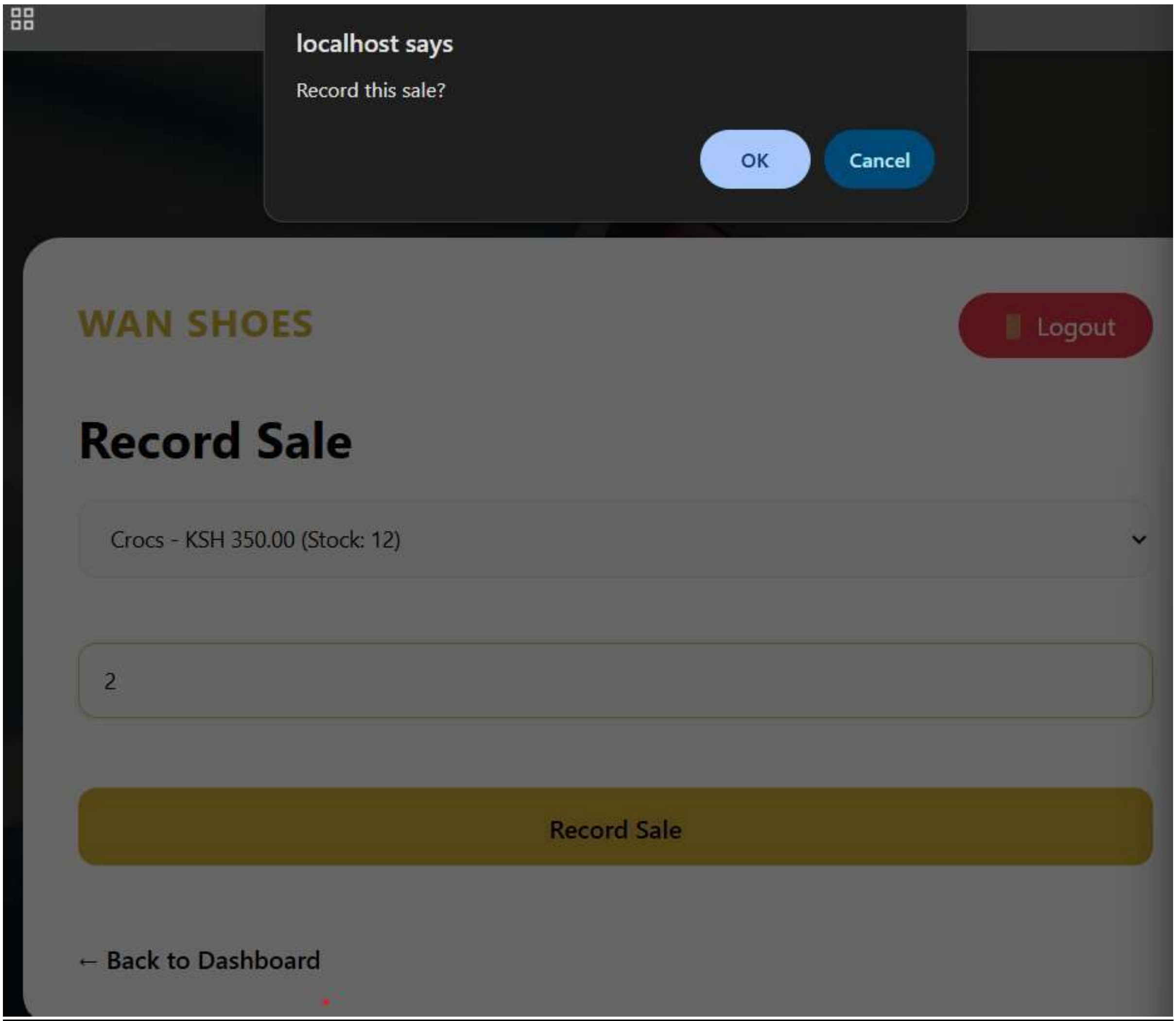


Manager Login


WAN
Welcome Back
Sign in to access inventory
manager@wanshoes.com
Sign In
Demo: admin@wanshoes.com / admin123


WAN SHOES
Logout
Employee Dashboard
Welcome, Jane Manager (Stock_manager)
Add Stock
View Products

Clerk Dashboard showing record sale

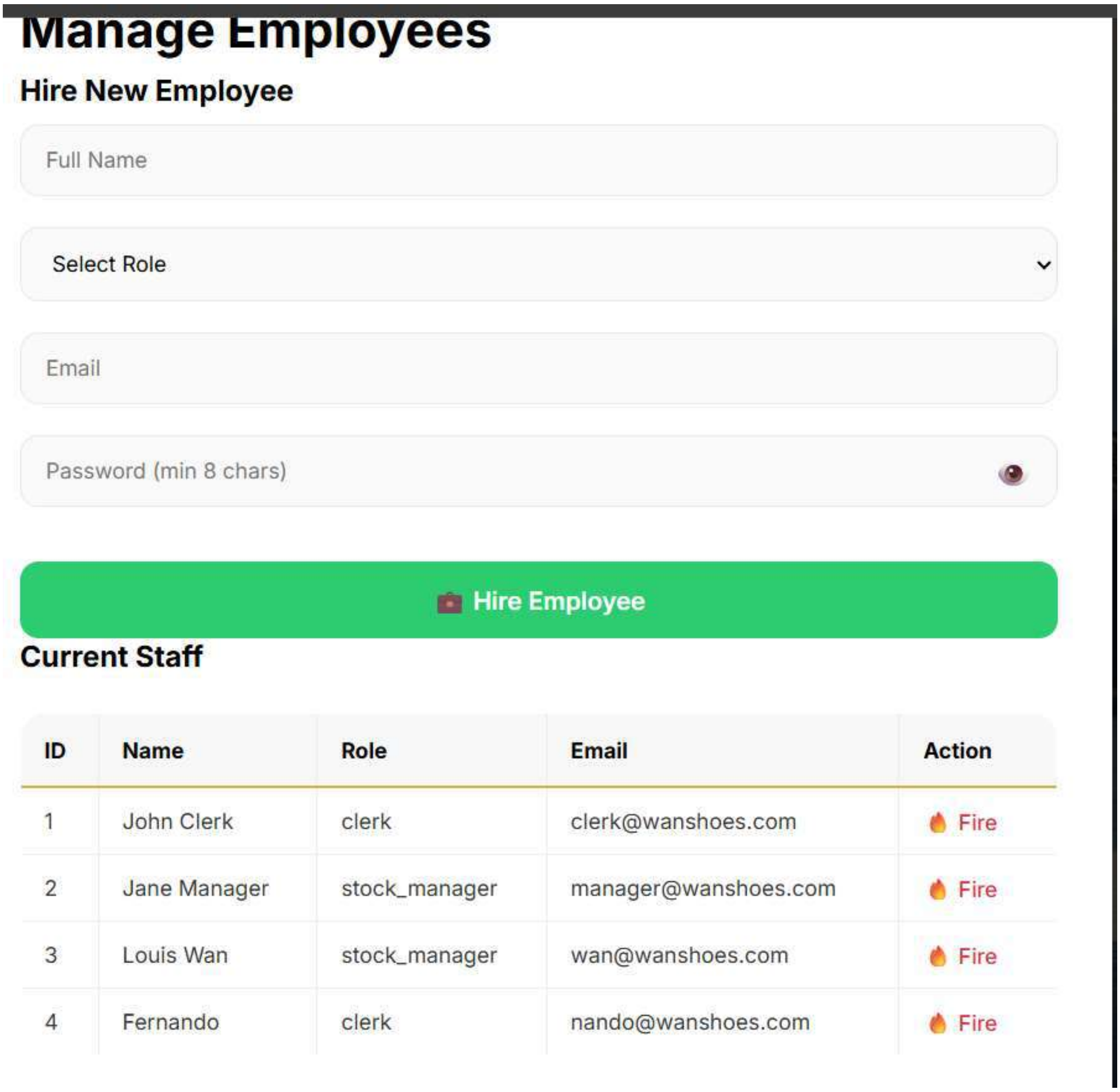


Student Reflection

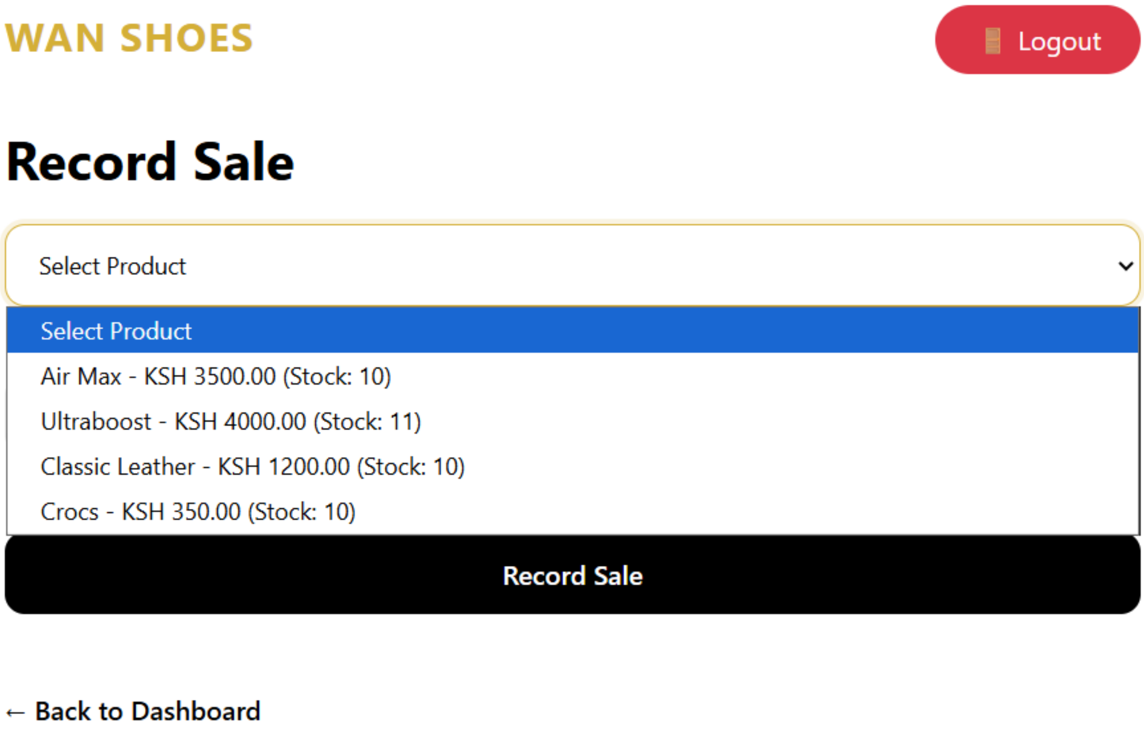
Week 4 focused on server-side authentication and session management for the Wan Shoes inventory system. I implemented a role-based login system that authenticates users against the admins and employees tables using mysqli_query(), then establishes sessions storing user_id, name, and role. The admin dashboard displays dynamic statistics calculated via SQL aggregate functions including SUM(stock_quantity * price) for inventory valuation showing total stock worth in KSH, and COUNT(*) for products and staff totals. Role-based redirection sends admins to admin.php while employees go to dashboard.php, where menu options differ for clerks (sales recording) versus stock managers (inventory addition). Session validation using isset($_SESSION['role']) is applied to all restricted pages, redirecting unauthenticated users to login.php. Logout functionality destroys sessions using session_destroy(). The folder structure maintains separation of concerns with CSS, JavaScript, and PHP controllers organized logically, demonstrating professional web development practices.

WEEK 5 Database Components and CRUD Operations

Hiring and Firing employees is only done by the admin



A clerk can view the products they can sell



An Admin can edit the products within the shop

Add New Product

Product Name

Brand

Size

Price (KSH)

Initial Stock

Add Product

Product List

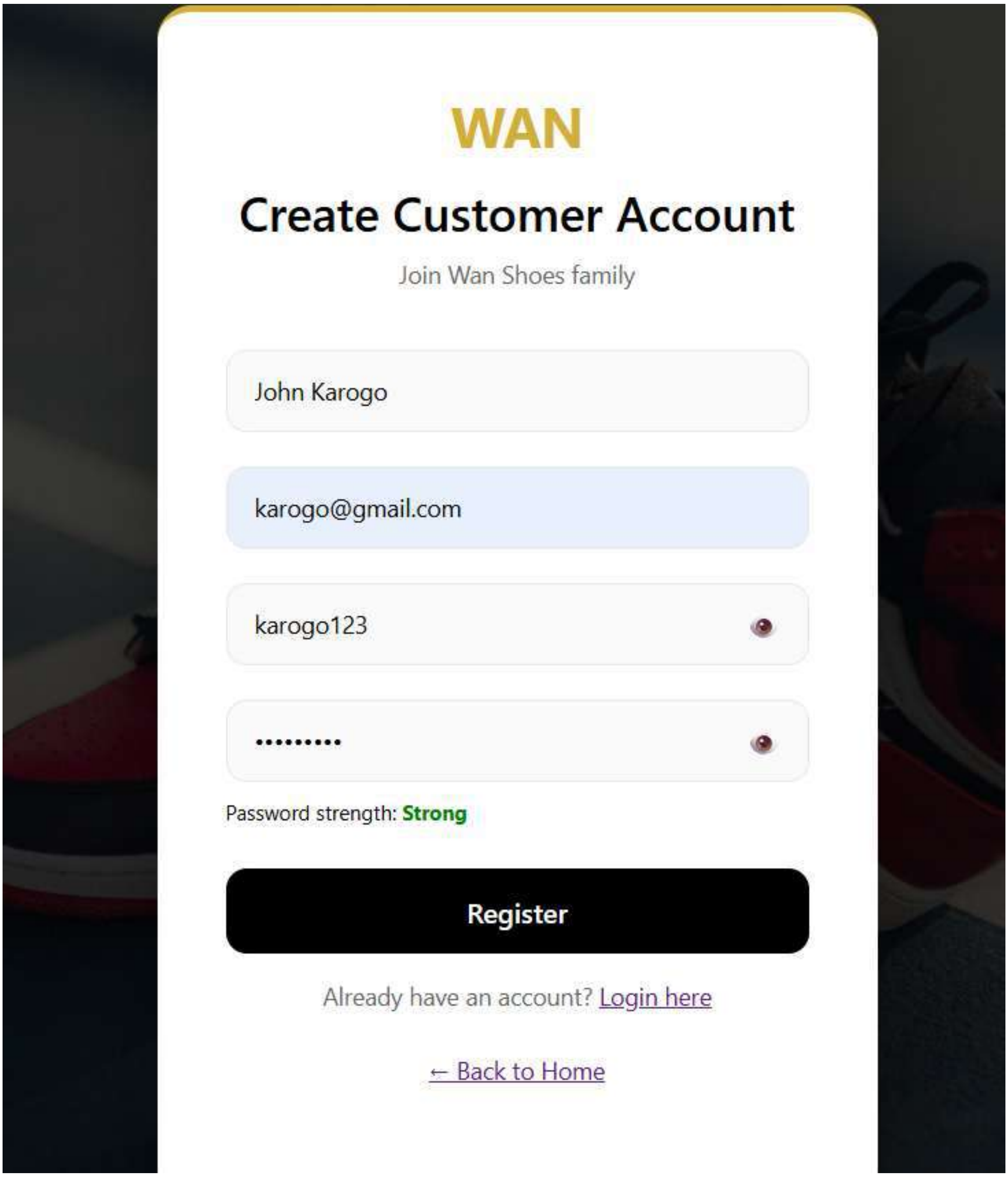
ID	Name	Brand	Size	Price	Stock	Actions
1	Air Max	Nike	42	KSH 3,500.00	10	Edit | Delete
2	Ultraboost	Adidas	41	KSH 4,000.00	11	Edit | Delete
3	Classic Leather	Reebok	43	KSH 1,200.00	10	Edit | Delete
4	Crocs	WANTAM	44	KSH 350.00	10	Edit | Delete

WEEK 5 REFLECTION

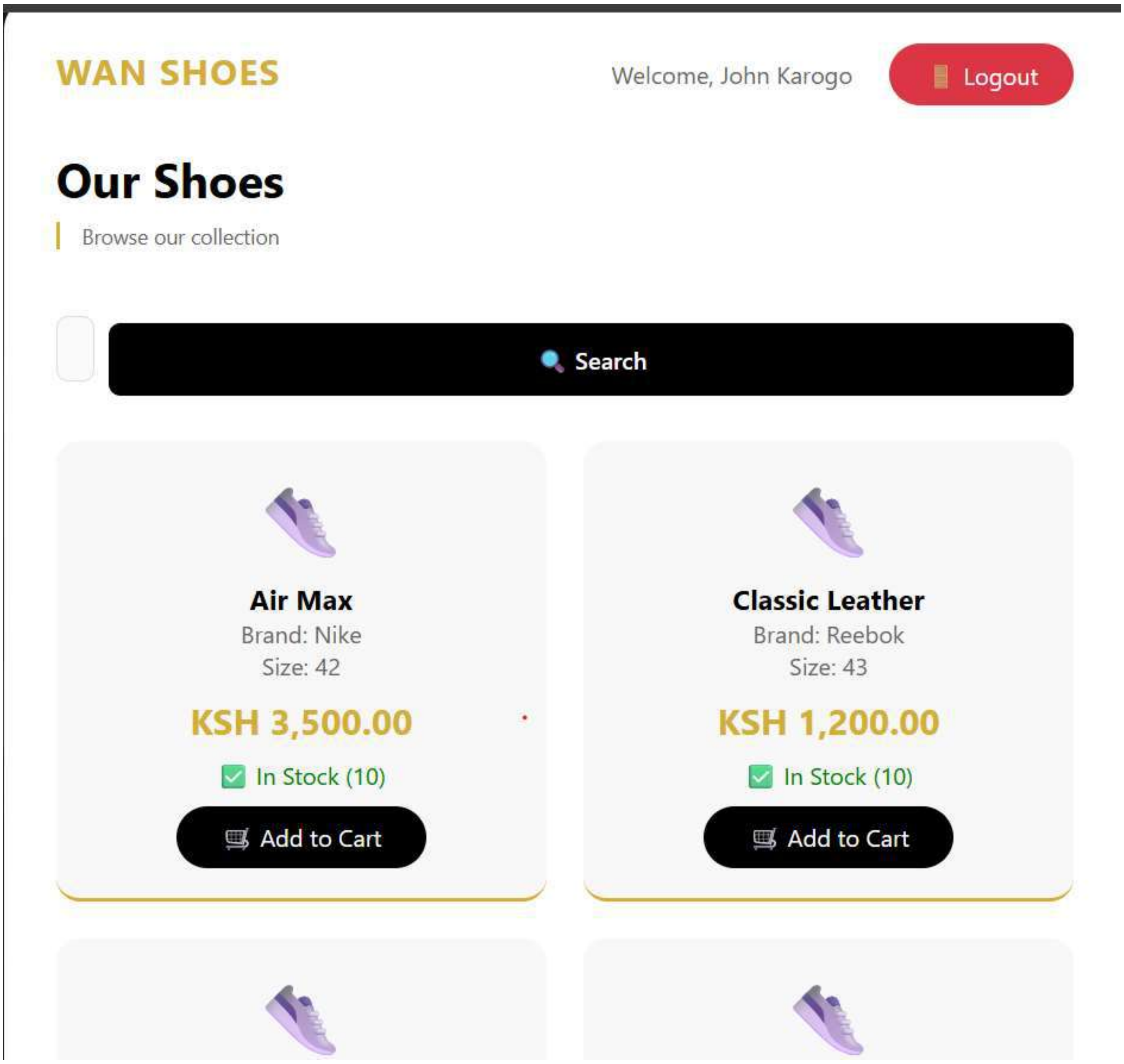
Week 5 completed the database-driven inventory system with full CRUD (Create, Read, Update, Delete) operations for the Wan Shoes platform. The products page allows admin to add new shoes with name, brand, size, price, and stock quantity, edit existing product details, and delete products — while clerks and stock managers have view-only access. Employee management enables admin to hire new staff with role selection (clerk or stock manager) and fire employees with a confirmation dialog. The sales recording page for clerks validates stock availability using a SELECT query before inserting into the sales table and updating the products table to deduct quantity. Stock managers can add inventory, updating product quantities and logging additions to stock_records for auditing. The admin sales report uses LEFT JOIN queries to display transaction details with clerk names, product names, and calculates total revenue using SUM() aggregation. All forms include JavaScript validation and the employee hire form includes a real-time password strength checker enforcing minimum 8-character passwords. The system is fully functional with all assignment requirements met.

WEEK 6: DATABASE INTEGRATION AND CRUD OPERATIONS

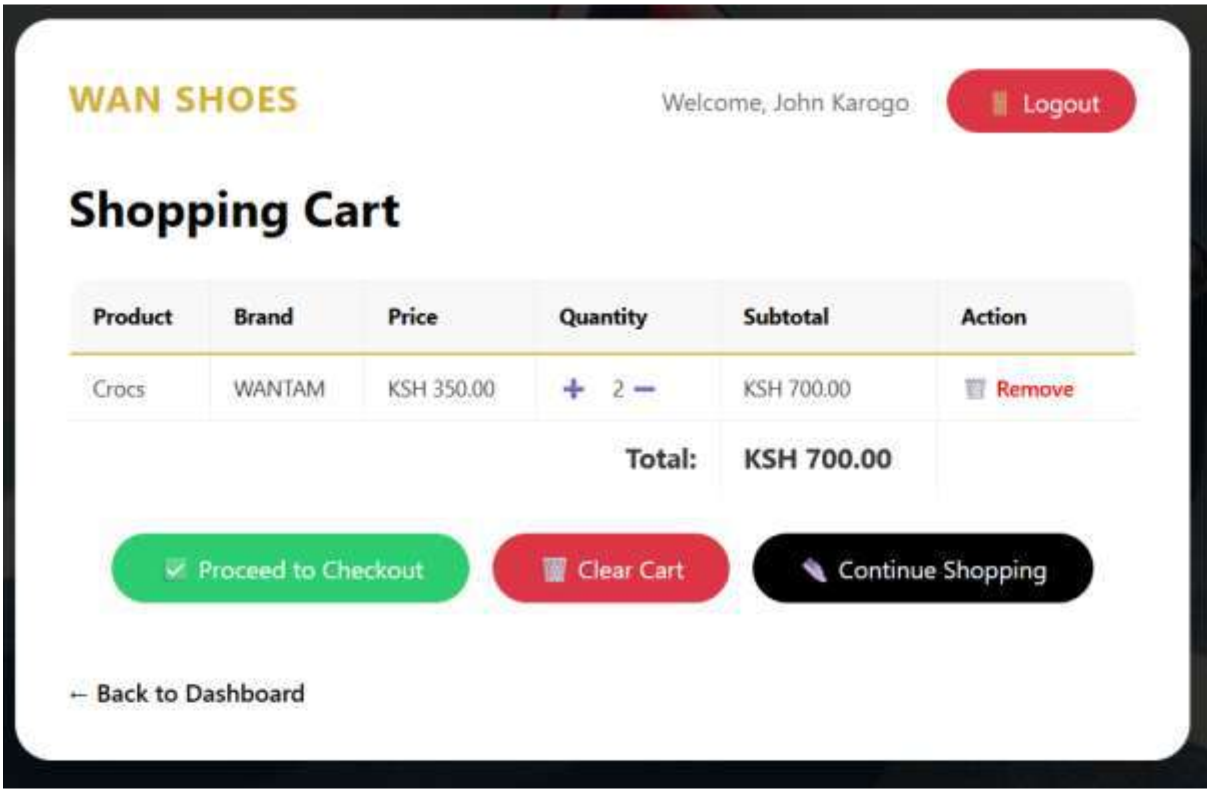
Customer registration system



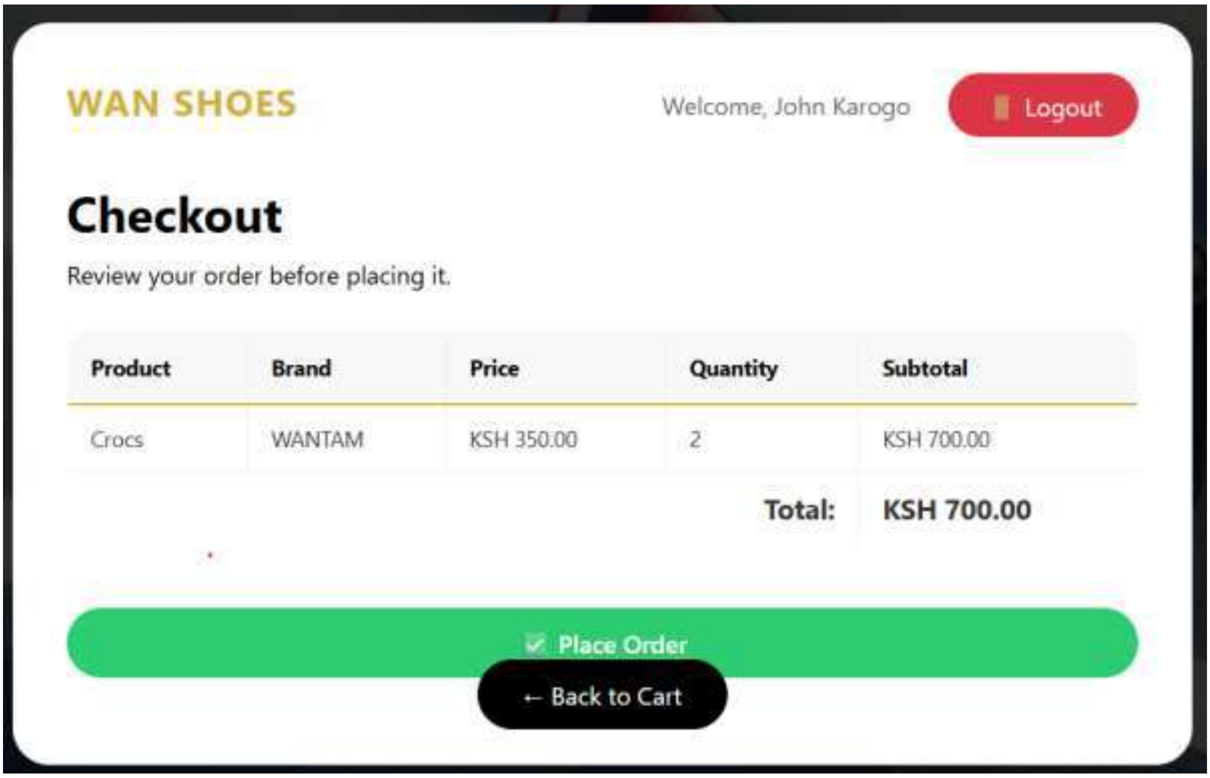
Customer Product Browsing



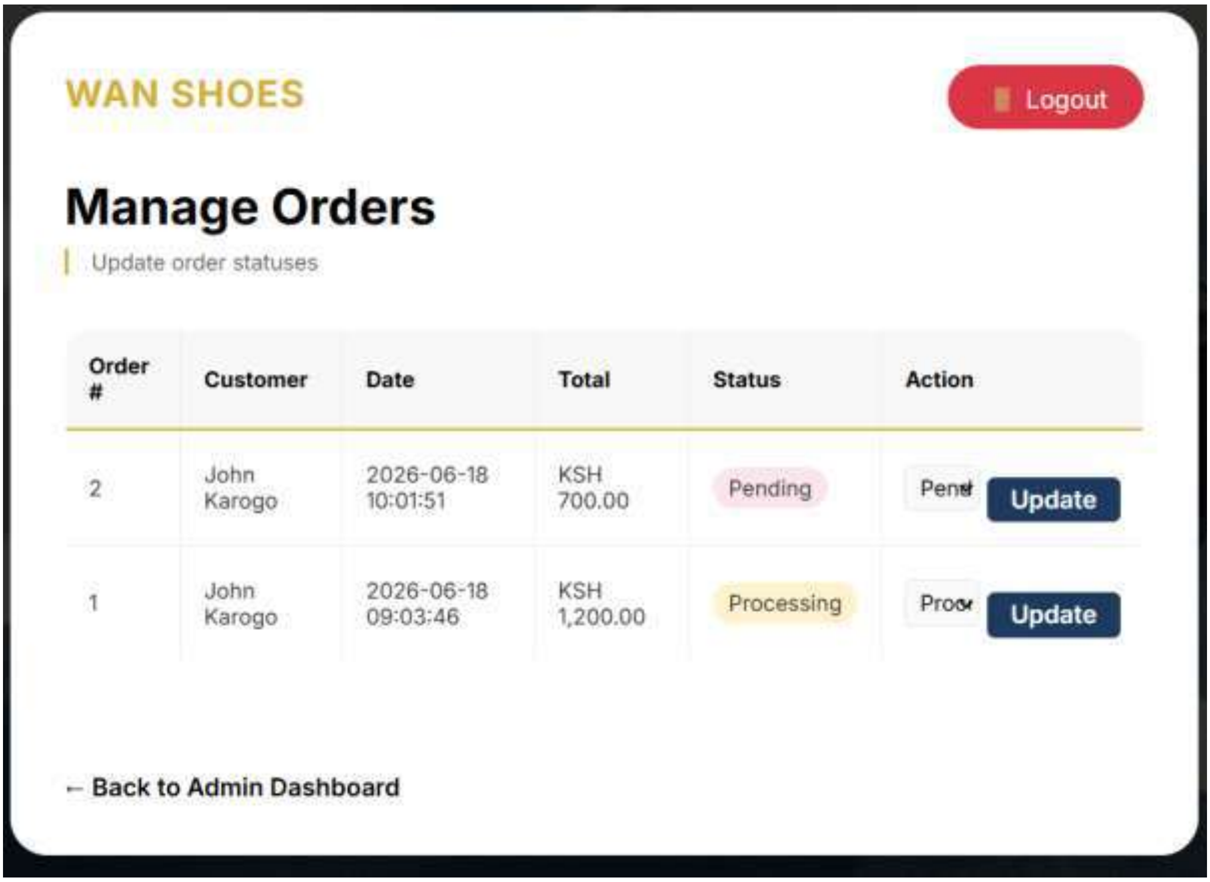
Shopping Cart



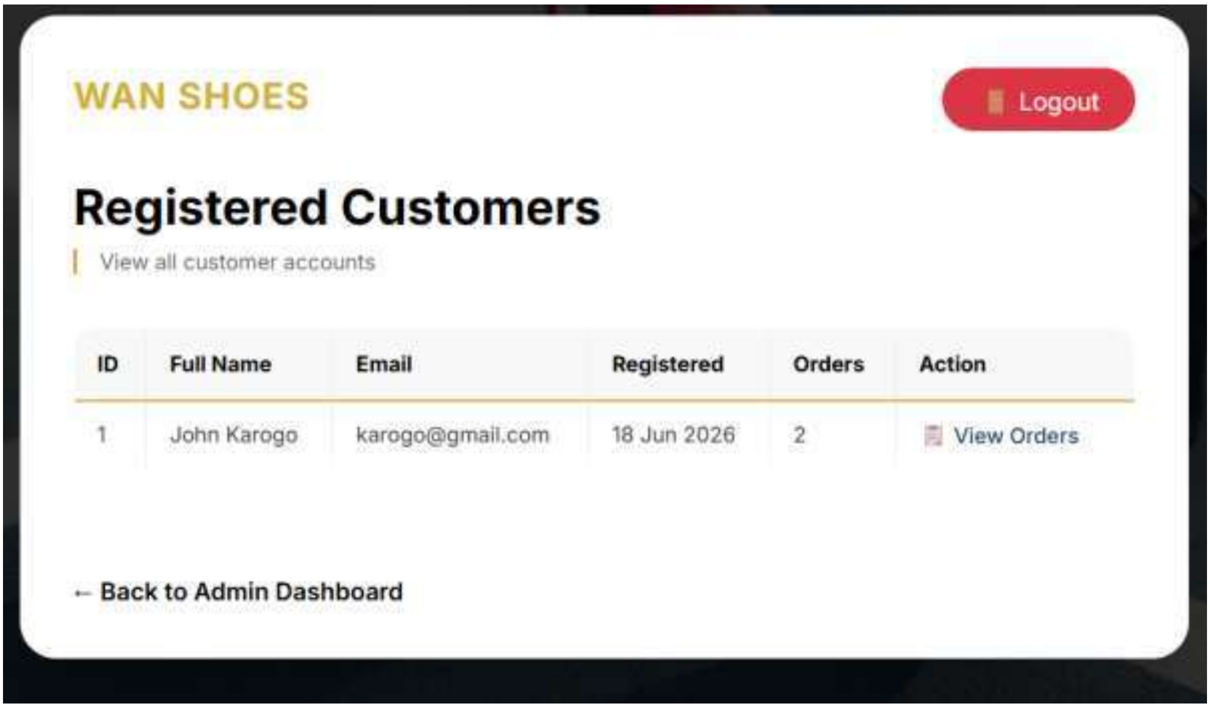
Checkout Page



Admin Manage Orders Panel



Admin Customers view

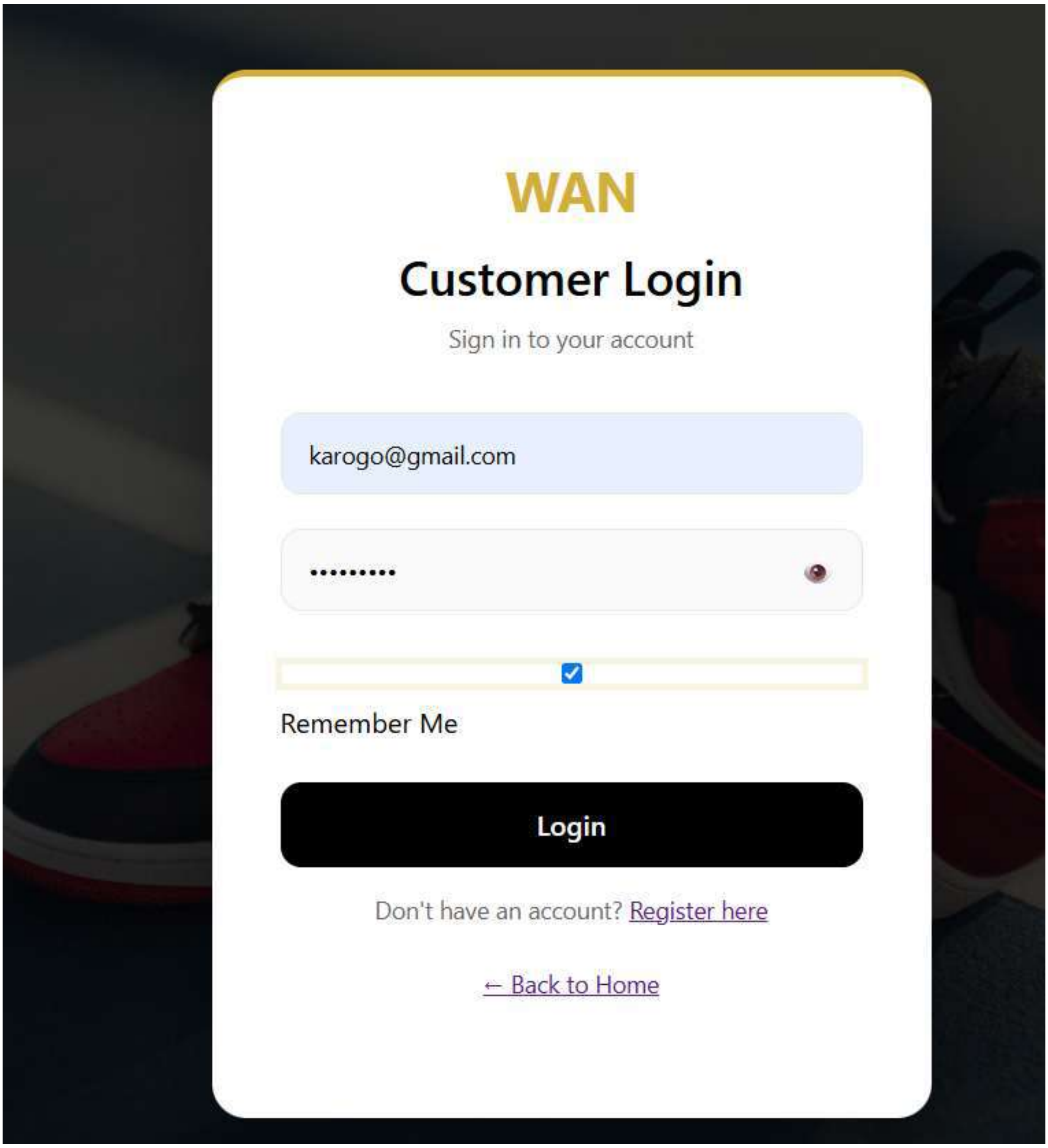


WEEK 6 REFLECTION

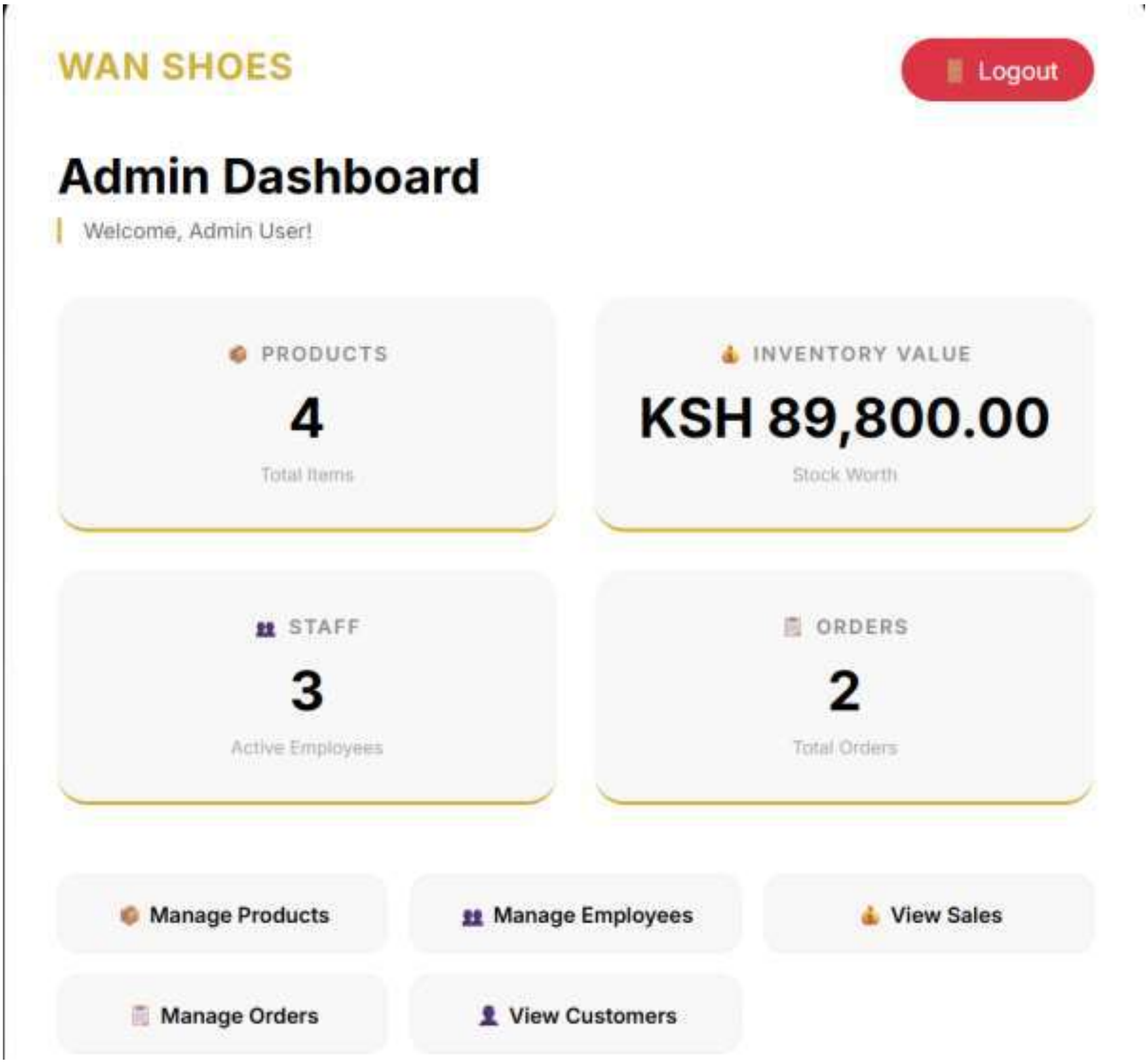
Week 6 extended the inventory system to include full customer-facing CRUD operations. Customers can register accounts, browse products with search functionality, add items to a session-based cart, and proceed to checkout. The checkout process creates orders and order_items tables, reducing product stock automatically. Admin can view all orders and update statuses (pending, processing, shipped, delivered) through a dedicated management interface. Sales reports now include a 7-day sales chart for visual revenue tracking. Customer management allows admin to view all registered users and their order counts. This complete CRUD cycle from customer registration to order delivery demonstrates a fully functional e-commerce system integrated with the existing inventory management.

WEEK 7 User Authentication and Session Management

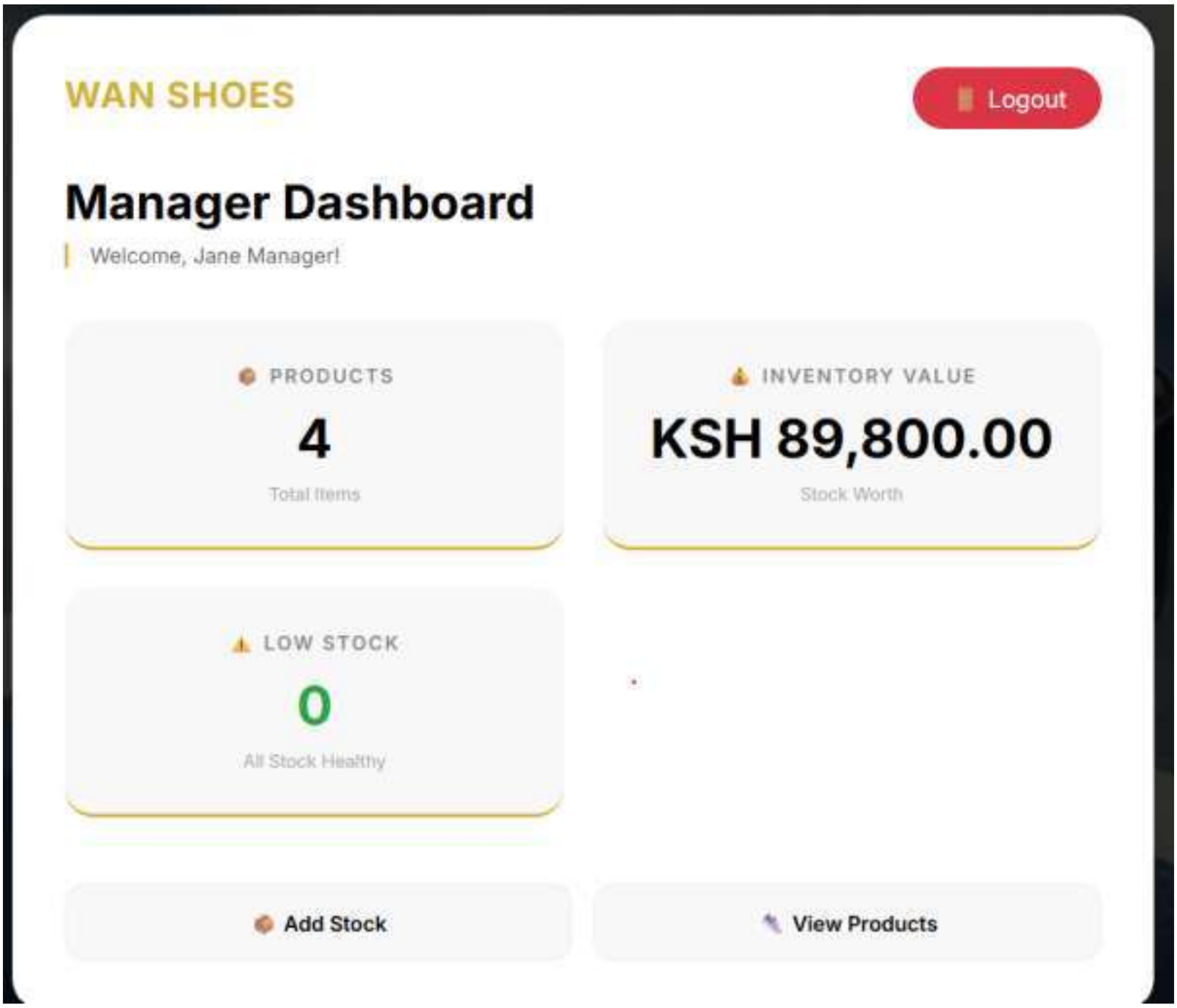
Login Page for Customers



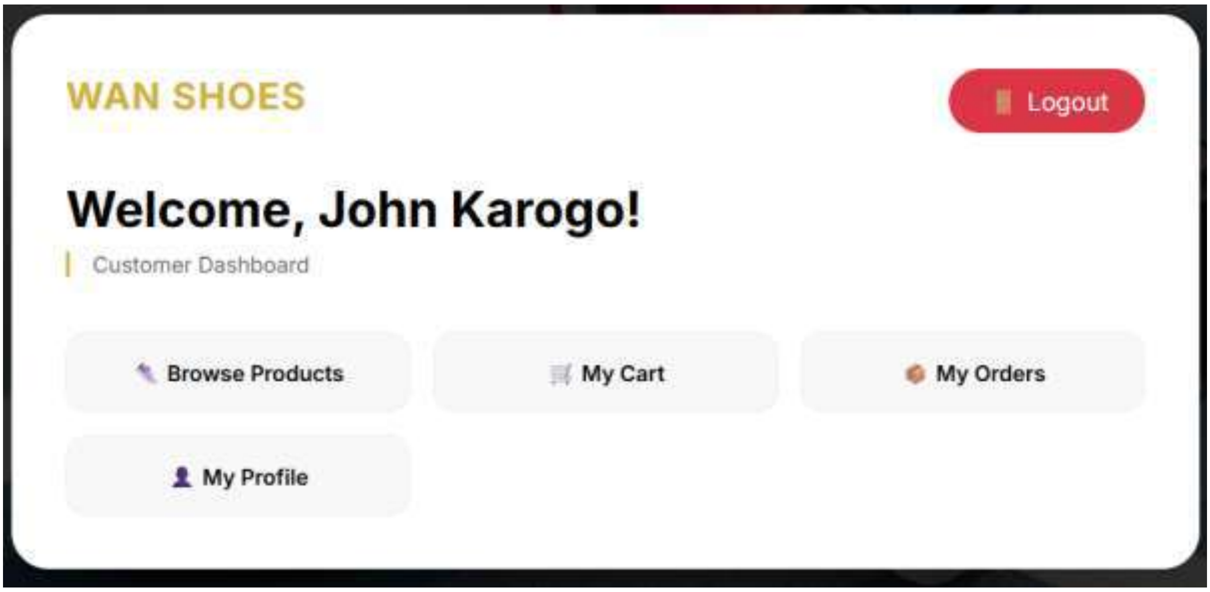
Admin Dashboard



Manager Dashboard



Customer Dashboard

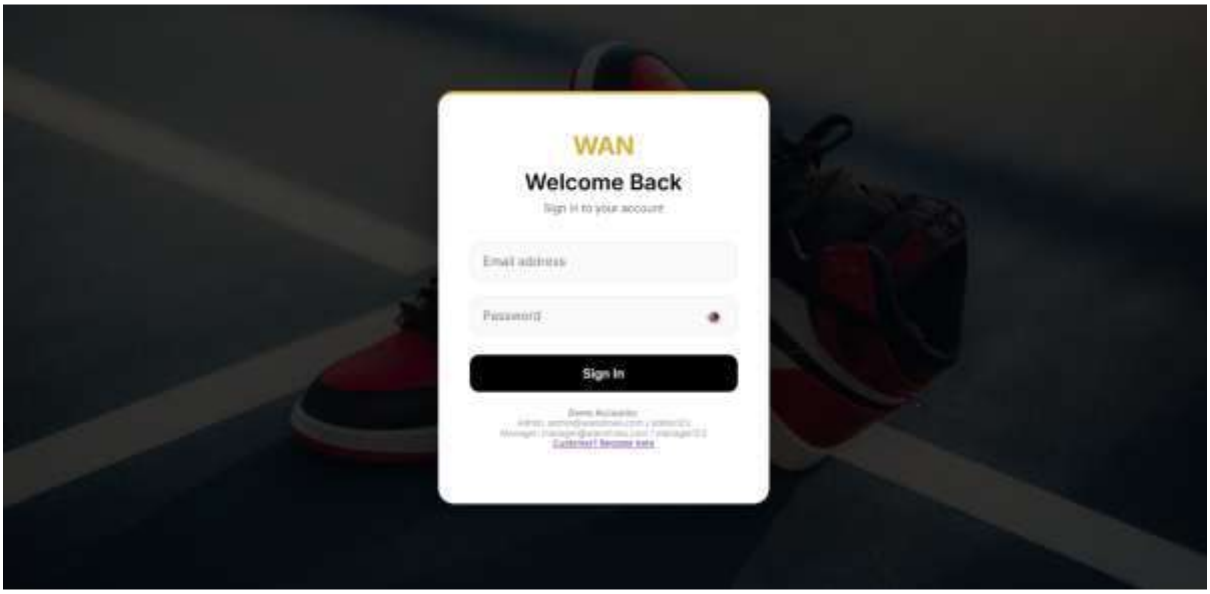


WEEK 7 REFLECTION

Week 7 implemented secure authentication and session management for the Wan Shoes system. User passwords are stored using password_hash() with bcrypt, and verified using password_verify() during login — eliminating plain text passwords. The login system distinguishes between Admin, Manager, and Customer roles, redirecting each to their respective dashboards. Session variables store user_id, name, and role for access control throughout the application. The "Remember Me" feature sets 30-day cookies for customer convenience, automatically logging returning users. Each dashboard includes session validation, redirecting unauthenticated users to login. Logout functionality destroys sessions and clears cookies. This authentication system meets all security requirements including proper hashing, session management, and role-based access control.

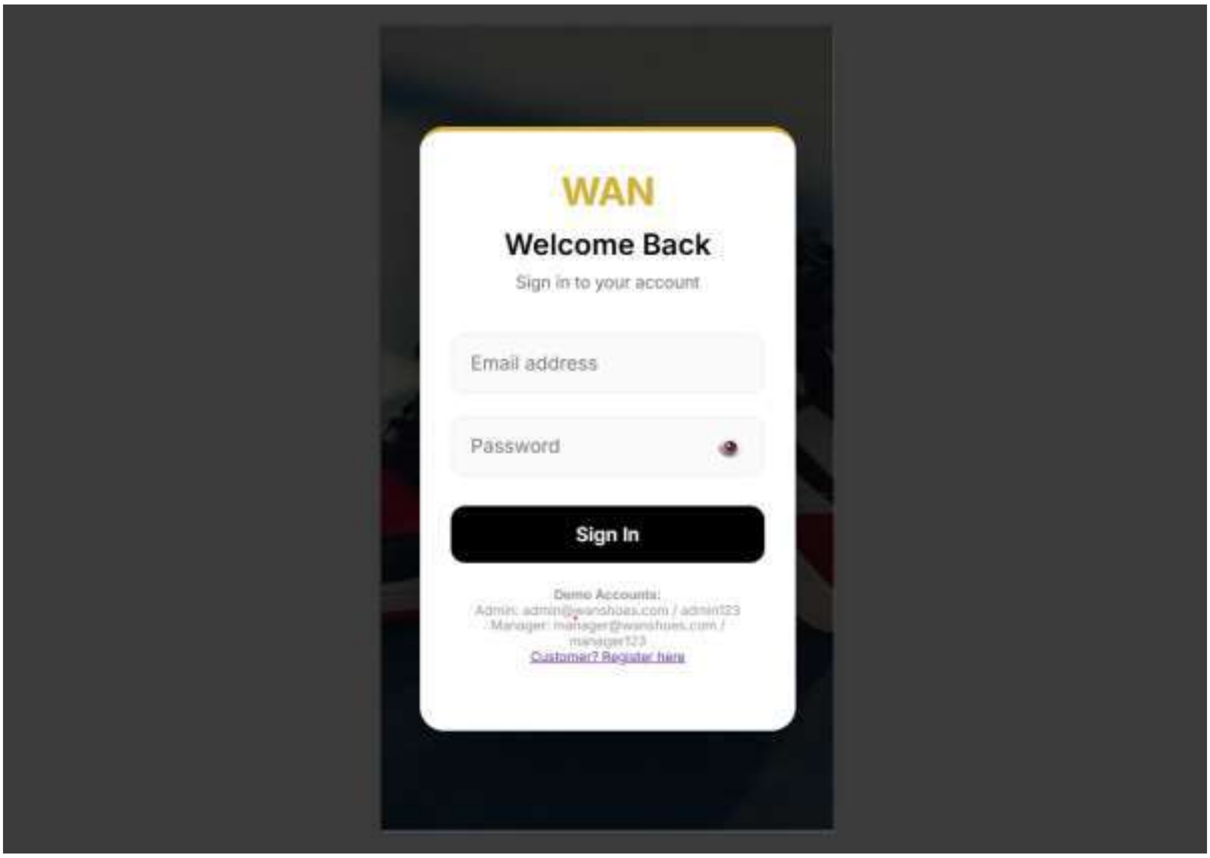
WEEK 8 RESPONSIVE WEB DESIGN AND MOBIE FIRST DEVELOPMENT

Desktop Login Page



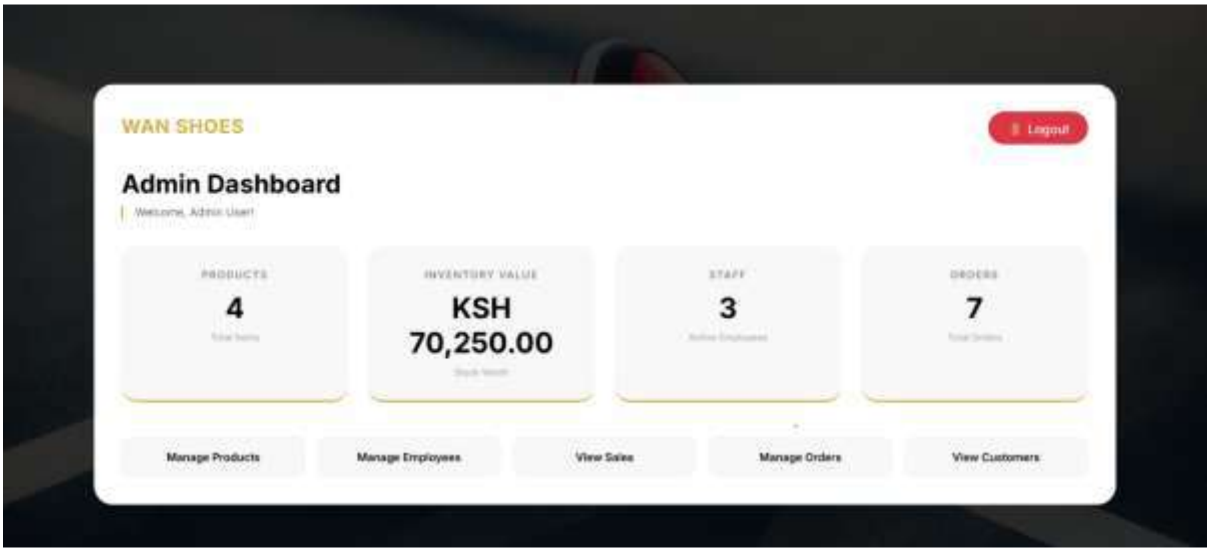
Desktop view of the login page showing full-width layout with centered container and form elements.

Mobile Login Page



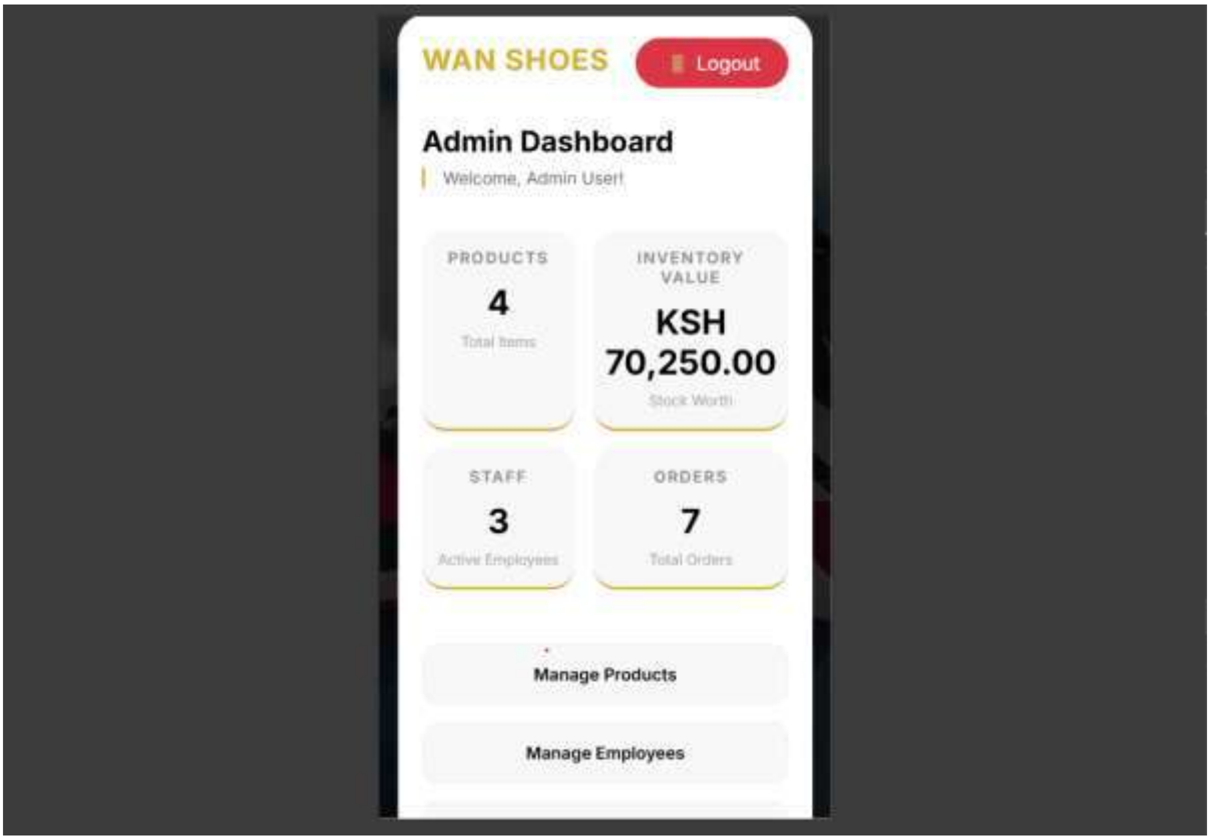
Mobile view of the login page on iPhone SE (375px) showing responsive container with proper padding and touch-friendly input fields.

Desktop Admin Dashboard



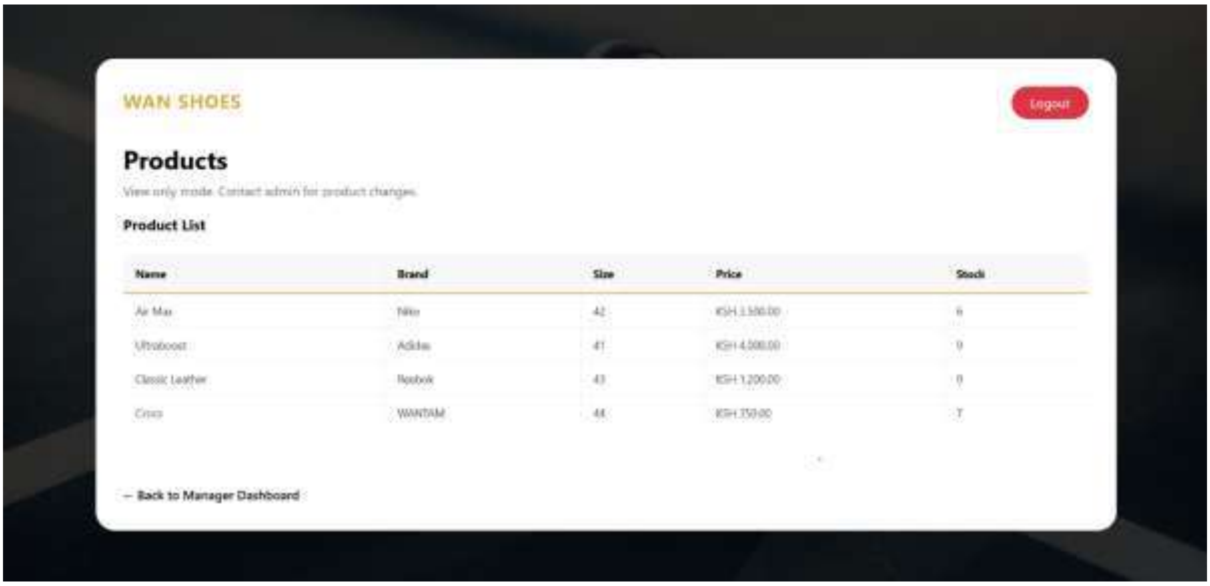
Desktop view of the Admin Dashboard displaying 5-column stats grid and full menu layout.

Mobile Admin Dashboard



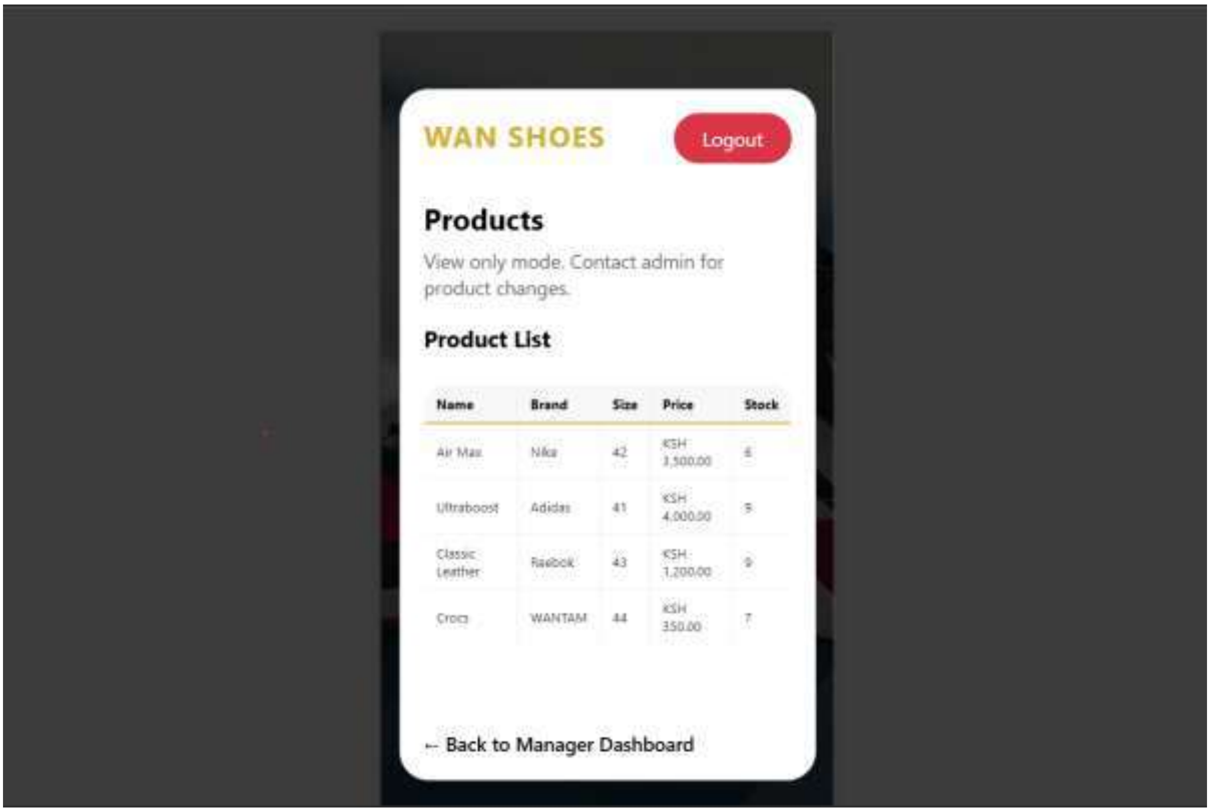
Mobile view of the Admin Dashboard on iPhone SE showing 2-column stats cards and single-column menu stacking vertically.

Desktop Products page



Desktop view of the products page showing full-width layout

Mobile Products page



Mobile view of the Products page showing responsive table with horizontal scroll on small screens.

Week 8 Reflection

Week 8 focused on implementing responsive web design using a mobile-first approach for the Wan Shoes inventory system. I added the viewport meta tag to all pages to ensure proper rendering on mobile devices. The existing CSS was enhanced with additional media queries for mobile (480px) and desktop (1024px) breakpoints. CSS Grid and Flexbox were used to create responsive layouts that automatically adjust from 1 column on mobile to 4 columns on desktop. Tables were wrapped with (.table-responsive) to enable horizontal scrolling on small screens, preventing layout breakage. Form inputs use 16px font size to prevent iOS zoom issues. Testing on Chrome DevTools confirmed the website works correctly on mobile, tablet, and desktop screens. These improvements ensure users can access the system comfortably from any device.